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1, The Platonic Solid

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The 7 stars of the Big Dipper are the 7 stars of the 7 Rishis.

Mizar and Alcor are two stars along the same line of sight. Alcor is used as a test for eyesight.

The 'toes' of the Bear are called 'the three leaps of the gazelle.'

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Surya Siddhanta, Chapter I with Commentary and Illustrations

Surya Siddhanta, from Surya, the Sun God, to Asuramaya, the Atlantean

Introduction

An adjustment of the previous translation from Sanskrit to English by Ebenezer Burgess, Surya Siddhanta, A Text Book of Hindu Astronomy, with notes and an appendix, 1858, originally printed in the Journal of the American Oriental Society, 1860. Presented here is chapter one.

Translation adjustments and comments by Alaska Mark. This retranslation is not accurate throughout. It is simply an adjustment of the Ebenezer Burgess translation. An acceptable translation would have the original Sanskrit in a phonetic spelling, followed by a word for word translation, followed by normal sentence structure in English. This for each of the verses. Here I have kept the small amount of Sanskrit words available in parentheses, accompanied by their English terms. These Sanskrit words and there English terms are in agreement with those used by the Sanskrit scholars of Theosophy, this that the language can remained standardized and all jargon can be avoided.

For greater accuracy one would have to reverse the whole sequence of translations. One could read Ebenezer Burgess's translation. One could compare that with other translations. One could learn to read Sanskrit and read a text from India. One could compare that with other versions found in India. Could one learn to read Atlantean and read a text there from? If one could then could one compare it with other texts from Atlantis? Could one learn to read the language of Asuramaya? If one could then could one read his Surya Siddhanta. Could one learn the language of a part of the Sun God (Surya)? If one could there still is the language of the Divinely illumined Surya who communes with the Gods. The Gods and Surya are with the reader, may the reader be guided by them while they sort through astronomical texts.

Quotations from Plato are from Timaeus, Benjamin Jowett translation, public domain.

Although presented here is only Chapter I the full text has the following table of contents:

Introduction
I. The Motions of the Planets
II. The Places of the Planets
III. Direction, Place and Time
IV. The Moon and Eclipses
V. The Sun and Eclipses
VI. The Projection of Eclipses
VII. Planetary Conjunctions

VIII. Of the Stars
IX. Risings and Settings
X. The Moon's Risings and Settings
XI. Certain Malignant Aspects of the Sun and Moon
XII. Cosmogony, Geography, and Dimensions of the Creation
XIII. The Gnomon
XIV. The Movement of the Heavens and Human Activity

One should read this book three or seven times. Once, to familiarize oneself with the themes and concepts of a Divine cosmology, twice, to see the interrelationships of those themes and concepts, and a third time, to put the themes and concepts into a holistic understanding.

So Asuramaya dwelt in the times of Atlantis and became renown as an astrologer and an astronomer. This astronomy was preserved through the record keepers of India, one document of which is said to be this Surya Siddhanta. The old Atlanteans of the Pacific island of Ruta traveled all the way to Egypt and had the same astronomy. The Greeks were from a more recent lineage of Atlanteans, from the Atlantic island of Poseidonis and there about, evidently, they went also into Egypt as well as to Athens, Greece. Their travels are recorded in Plato's Timaeus. These Atlantean Egyptians and Greeks also had this astronomy. I have found agreement between the Timaeus of Plato and the Surya Siddhanta.

All of the comments and calculations are mine. They are subject to question and examination and any corrections can be added as needed. This is an attempt to understand the Siddhanta by exploring the possible calculations. I've started with the assumption that the Surya Siddhanta is correct and that it agrees with Plato's Timaeus. It has been up to me, the student, to try to understand the math of Asuramaya, Surya, and indeed, of Brahma for the numbers and geometry were used during the time of creation. To do this I have had to put aside a lot what is popularly accepted and adhere to what is true only. Story has it that Asuramaya worked out extensive patterns, from astronomy well into astrology and the evolutionary times of humans, the seven root races, and the courses planets and stars. The reader is encouraged to work out the calculations for themselves, for two reasons. One, there is a considerable boon to anyone who calculates the sum of days, and two, to see if my calculations are correct or incorrect.

The translation is the same as the preceding. The comments are such as to increase the intermediary understandings. The calculations are kept short and within the ability of the average reader. Universal design is based on what the west calls Pythagorean number and sacred geometry. Added in parentheses are what Sanskrit words I have available. I am not an experienced Sanskrit scholar so the reader is advised that English phonetic spellings are slowly become standardized and that there are metaphysical meanings to each word.

Chapter I

The Motions of the Planets

1. To Him whose shape is inconceivable and unmanifested, who is unaffected by the qualities, whose nature is quality, whose form is the support of the entire creation, to Brahma be homage.

Comment: From the Timaeus:

Timaeus: All men, Socrates, who have any degree of right feeling, at the beginning of every enterprise, whether small or great, always call upon God. And we, too, who are going to discourse of the nature of the universe, how created or how existing without creation, if we be not altogether out of our wits, must invoke the aid of Gods and Goddesses and pray that our words may be acceptable to them and consistent with themselves. Let this, then, be our invocation of the Gods, to which I add an exhortation of myself to speak in such manner as will be most intelligible to you, and will most accord with my own intent.

2. When but little of the Golden Age (Krita Yuga also called Sathya Yuga) was left, a great asura named Asuramaya, being desirous to know that mysterious, supreme, pure, and exalted science,

Comment: Asuramaya as in asura, the lower earthly counterparts whom are good and evil. The upper, the Gods, are sura. Maya means appearance.

Shortly after the time referred to would be the end of the last Golden Age and the start of the Silver Age. The ages in succession being Gold, Silver, Bronze, and Iron. In India these are called Sathya Yuga, Treta Yuga, Dwapara Yuga, and Kali Yuga. Sathya Yuga is also called Krita Yuga. These ages are in the proportion of 4 to 3 to 2 to 1.

The Greek Tetraktys

We note that 4 + 3 + 2 + 1 = 10. 1 part in 10 equals 432,000 years and all 10 parts are a Mahayuga equal to 4,320,000 years. Therefore looking at our current Mahayuga:

Shortly before the end of the Golden Age (Sathya Yuga), 4 x 432,000 = 1,728,000 a few years

and before the start of the Silver Age (Treta Yuga), 3 x 432,000 = 1,296,000 1,296,000

Bronze Age (Dwapara Yuga), 2 x 432,000 = 864,000 864,000

Iron Age (Kali Yuga) 1 x 432,000 = 432,000 5,104

total 2,165,104

The times of Asuramaya seem to be about 2,165,104 years ago.

So when did the Iron Age (Kali Yuga) start? The wise men of India say the Kali Yuga began when Krishna went to his heavenly abode in 3102 BC., February 17 or 18. If Krishna left February 17 then the Kali Yuga is thought to have started on the 18'th. I still wonder why the equinox isn't mentioned. Another story says that a deer hunter had inadvertently shot Krishna in the toe with an arrow while hunting in the forest and then Krishna went on to Krishnaloka, world of Krishna. There are many deer hunter and the wise man stories in India. Orion is called Mriga, the antelope headed.

From 3102 BC till 2003 AD is 5,104 years. The Kali Yuga is our present yuga and continues until 428,897 AD. The time referred to is a little more than 2,165,104 years ago. These were the times of Atlantis, I would think, even before the time of Rama and Hanuman.

We note that January 1, 1 BC to January 1, 1 AD is 1 year, not 2. 10 BC to 10 AD is 19 years,

not 20. 3,102 added to 2,003 is 5,105. Therefore 3102 BC to 2003 AD is 5104 years, not 5105.

3. That chief auxiliary of the scripture (Vedanta), in its entirety, the cause, namely, of the motion of the heavenly bodies (jyotis), performed, in propitiation of the Sun God (Surya), very severe religious austerities.

Comment: Another of Surya's titles is Surya Narayana. This translates as The Divine Light of the Sun God, where Narayana means the original spirit that moved across the casual ocean. Another name for the Sun God is Ravi The other planets (grahas) with their Hindu names are the deities of the moon, (Soma and Chandra), Mars (Mangal, also called Kuja), Mercury (Budha), Jupiter (Guru, which means transmitter of knowledge), Venus (Sukra), and Saturn (Shani). Chandra, the lunar goddess, is said to dwell in the mind and is considered omnipresent and yet mind is said to be the moon. Surya the Sun God is called Helios in the Greek. Later Helios was called Apollo but this isn't correct for Apollo is a the god of the muses, also called Musagetes.

Surya the Sun God, called Helios in the Greek.

4. Gratified by these austerities, and rendered propitious, the Sun God (Surya) himself delivered unto that Asuramaya, who besought a boon, the system of the planets.

The blessed Sun God (Surya) spoke:

5. Your intent is known to me; I am gratified by your austerities; I will give you the science upon which time is founded, the grand system of the planets.

6. No one is able to endure my brilliancy; for communication I have no leisure; this person, who is a part of me, shall relate to you the whole.

Some texts have here:

Go therefore to Romaka City, thine own residence; there, undergoing bodying forth as a barbarian, owing to a curse of Brahma, I will impart to you this science.

Comment: Bodying forth would be the Sun God (Surya), manifesting a part of himself to dwell among mortals. India speaks of the three worlds, earth, (Bhumi or Bhu), mid heaven (Bhuva), and high heaven (Suvaha). The Sun God (Surya) exists in high heaven (Suvaha) where is the city of Brahma (Brahmaloka), the gods (sura) and the origin of the creation. The Sun God (Surya) exists in mid heaven (Bhuva) where are the stars (bha), the Sun, the moon, and the planets (grahas), as well as their numbers of revolutions in proportion to one another. The Sun God (Surya) can exist on the earth (Bhumi) as a bodying forth, a type of avathara, to dwell among mortals. Avathara means Divine appearance as in the Avatharas of Vishnu or Shiva. For the Sun God (Surya) this has been reported to have happened at other times, one of

which is his appearance among the combatants on a Greek battlefield, another of which is his appearance in Peru near Lake Titicaca.

I myself think that Hermes is a bodying forth (type of avathara) of Mercury (Budha). Another example is Padma Sambhava as a bodying forth (type of avathara) of the Buddha Amitabha. Padma Sambhava is credited with the writing of the Bardol Thodal, the Tibetan Book of the Dead.

A list of ten of the avatharas, evidently in the order of their appearance, of the Divine are; one, Matsya, the fish avathara; two, Kurma, the tortoise avathara; three, Varaha, the boar avathara; four, Narasimha, the Divine man lion avathara; five, Vamana, the dwarf avathara that was a giant; six, Parasurama, the warrior avathara.

Then seven, Rama, the defeater of the demon king Ravana of Lanka during the times of Atlantis; eight, Krishna, the flute player and teacher of the cow herd boys. Krishna left this worldly scene February 17, 3102 BC. Nine, the triple avathara of today, Shirdi Sai, 1835 to 1918, the peace maker; Sathya Sai Baba, 1926 till today, the great world teacher; and Prema Sai, expected in 2029 or 2030. And ten, Kalki, the avathara yet to appear at the end of this Iron Age (Kali Yuga) and the start of the next Golden Age (Sathya Yuga). The times of the avathara is the way India regards history. A million years from now not much will be remembered of our times but the name of Sai Baba will be remembered.

The major world centers at the time of Atlantis.

Romaka was one of five principle cities or centers of the time, and indeed today, for they are spiritual centers, the five being:

1. Siddha, the northern city of the land of Kuru, today above the north polar region;

2. Lanka, southern city of the land of Bharata, today as India. This phonetic continues in the name of the island of Ceylon as Sri Lanka. Near Lanka was Rohitaka.

3. Yamakoti, the eastern city of the land of Bhadrasva, today south of Wake Island and Bikini Atoll;

4. Romaka, the western city of the land of Ketumala, today off the coast of West Africa;

5. Patala, 'the other side', today Peru is on the other side of the earth from India, and Guadalajara, Mexico, the land of Quetzaquatal, is on the other side at the same latitude.

These centers are on the quadrature from one another and seem to be more spiritual centers than just physical cities. The city of Ujjain, India has been regarded as the prime meridian for ages. The world grid, both at the time of this Siddhanta and now, would be placed with

Ujjain to the right, south;

Yamakoti, today as south of Wake Island and Bikini Atoll, to the front, east;

Romaka, today off the west coast of Africa, to the back, west, this phonetic seems to continue in the name 'Rome';

and Patala, today as Peru and Guadalajara, to the left, ‘the other side, Peru as directly opposite and the area around Guadalajara as the opposite side at the same latitude. Near Guadalajara to the east is Teotihuacan.’

Giza, Egypt, then, would be at, or nearly at, 45 degrees west of the prime meridian of Ujjain, India. North of Ujjain in the land of Kashmira is the town of Srinagar. This town is said to be associated with the Hebrews and the travels of Jesus.

A world grid with principal cities, where Romaka has been called the ‘city of the sun.’

7. Thus having spoken, the Sun God (Surya) disappeared, having given directions unto the part of himself. This latter person thus addressed Asuramaya, as he stood bowed

forward, his hands suppliantly joined before him:

8. Listen with concentrated attention to the ancient and exalted science, which has been spoken, in each successive age, to the Great Sages (Maharishi), by the Sun God (Surya) himself.

Comment: A lengthy commentary on Rajarishis and Maharishis, Manus and emanations.

From what I've read I've almost convinced myself that this can be properly presented. I still wonder if the 7 Rishis are the 7 Manus or if the Manus followed after the Rishis. Here I present it several ways to familiarize myself with the nomenclature and the system. The information is from the Vedas of India and the Surya Siddhanta and the Theosophy of Madame Blavatsky.

Vishnu is the sustainer of universes and Brahma is the creator of this universe. The 7 Rishis are Rajarishis, from Raja, royal and rishi, scribe. The 7 Rishis are emanations of Brahma.

This is the 28'th Mahayuga of the Manu Vaivasvata, son of Vivasvant (Vivaswan).

Manu, is the son of Brahma and the progenitor of Mankind. The Manus are 7, which is 14. The 7 in turn emanate 7 for 49, and from these 49 come the progenitors of the 7 root races of our humanity. The arrangement is called emanations.

Vishnu

Brahma

The 7 Rajarishis

Manu

The 7 Manus which are 14

The 7 emanations of the 7 Manus for 49 emanations

The 7 root races of humanity

The myriad generations

From the first Manu there are 7 emanations which are listed as 14 Manus for 1 kalpa, the day of Brahma, the time for this creation. We are at the 7'th of 14 with 6 before this one and 7 after this one. Theosophy gives their names as follows:

Swayambhuva

Swarochi or Swarotisha

Uttama

Thamasa

Raivata

Chackchuska

Vaivasvata, son of Vivaswan, the current Manu, we are in the 28'th Kali Yuga

Savarna

Daksha Savarna

Brahma Savarna

Dharma Savarna

Rudra Savarna

Rouchya

Bhoutya

www .vahini. org, glossary gives the spelling of the 14 Manu's as Svayambhuva, Svarocisa, Uttama, Tamasa, Raivata, Caksusa, Vaivasvata, Savarni, Daksa-savarni, Brahma-savarni, Dharma-savarni, Rudra-savarni, Deva-savarni, and Indra-savarni. Elsewhere Vaivasvata is also spelled as Vaivaswatha.

The current Manu is Vaivasvata son of Vivaswan. Each Manu predominates for a manvantara which is 70 and 1 Mahayugas and a twilight. Mahayuga is Maha, great and yuga, age. This is the 28'th Mahayuga of this manvantara. One Mahayuga is 4,320,000 years.

So today's date would be 2004 AD, the 5,105 year of the 28'th Kali Yuga, of the 7'th manvantara, the manvantara of the Manu Vaivasvata, son of Vivaswan, the 7'th of the 14 Manus of Brahma, in this kalpa.

1 day of Brahma = 1 kalpa, the time for this creation.

1 kalpa = 14 manvantaras each with their respective twilights

1 manvantara = 70 and 1 Mahayugas and a twilight

The years can be expressed in Divine years or solar years with a ratio of Divine years / solar years = 1 / 360

In solar years:

1 twilight = 1,728,000 years

1 Mahayuga = 4,320,000 years

1 Mahayuga = 1 Sathya Yuga + 1 Treta Yuga + 1 Dwapara Yuga + 1 Kali Yuga

1 Sathya Yuga = 4 x 432,000 = 1,728,000 years

1 Treta Yuga = 3 x 432,000 = 1,296,000 years

1 Dwapara Yuga = 2 x 432,000 = 864,000 years

1 Kali Yuga = 1 x 432,000 = 432,000 years

The 7 Rishis are Rajarishis and also Maharishis, from Maha, great and rishi, scribe. The 7 Rajarishis coexist with the 7 Maharishis as the foundation of spirit and mind. The 7 Maharishis are Bhrigu, Marichi, Atri, Angira, Pulastya, Pulaha, and Kratu. It seems to me that the Maharishis are from the emanations:

Vishnu

Brahma

The 7 Rajarishis

Manu

The 7 Manus which are 14

The 7 emanations of the 7 Manus for 49 emanations

The 7 root races of humanity

The 7 Maharishis, Bhrigu, Marichi, Atri, Angira, Pulastya, Pulaha, and Kratu

The myriad generations

India has the temple of Bhrigu where their book is written on actual leaves. People go to the temple and consult with the priest on matters of wisdom, and the priests sort through the leaves to find the proper set of writings. The calculations of Bhrigu are said to be amazing. Can one know how many leaves will grow on a tree, and when they grow and mature and change colors in the autumn, and can one then say where each leaf will fall?

There is a design to the flow of books, and leaves of books, into and out of each person's life.

The same can be expressed in the Jewish system of Kabbalah as the doctrine of emanations:

Vishnu, Ein Sof

Brahma, Ein Sof, the unlimited light

Ein Sof An, the limited light

Ein Sof Or, creativity

The 7 Rajarishis, the 7 Elohim, first emanation

Manu, Adam Kadmon, cosmic man

The 7 Manus which are 14, the 7 times

The 7 emanations of the 7 Manus for 49 emanations, the 7 times 7

The 7 root races of humanity, Adam Kadmon and the universal plan for man

The created humanity, Adam and Eve

The 7 Maharishis, Moses and the great scribes

The myriad generations, the 12 tribes of Israel

The 7 part wisdom of emanations then includes the Rajarishis, the Maharishis, the 7 stars of the Big Dipper, the 7 muses of wisdom, and the 7 stars of the Pleiades.

The 7 stars of the Big Dipper are the 7 stars of the 7 Rishis.

Mizar and Alcor are two stars along the same line of sight. Alcor is used as a test for eyesight.

The 'toes' of the Bear are called 'the three leaps of the gazelle.'

Ursa Major, the bear, called Arktos by the Greeks. They also called it Ixion, the wagon that goes around the axis mundi.

Dubhe Arabic, from **Al Thahr al Dubb al Akbar**, the back of the bear the greater. **Dubhe** is also called the thigh.

Merak Arabic, from **Al Marakk**, the loin.

Phecda Arabic, from **Al Fahdh**, the thigh.

Megrez Arabic, from **Al Maghrez**, the root of the tail.

Alioth From the Alphonsine Tables of 1252 or 1437 or both.

Mizar Arabic, from **Mi'zar**, girdle or waist cloth.

Alkaid Arabic, from **Al Ka'id**, from **Al Ka'id Banat al Na'ash**, the governor of the daughters of the mourners or the governor of the daughters of the square like geometry. The square like geometry is the four stars of the dipper part and seems to be a reference to square one, that square used by the creator. This geometry is call **Bier** in the Arabic, **Pih Tow** in the Chinese, and **Ash** or **Ayish** in the Hebrew. The square or near square seems duplicated 180 degrees around the circle in the square of Pegasus and again in the little dipper of **Sagittarius**.

Alcor, next to **Mizar** so that a person of keen eyesight would notice two stars where others had but counted one. **Mizar** and **Alcor** have been a test for eyesight among sailors of the north and Arab al Baida of the south; the Arabs of the desert.

Muscida, muzzle, the nose of the bear.

Hewitt is referred to by Richard Hinckley Allen as writing that the 7 sages, the Rajarishis, assigned to the 7 stars of **Ursa Major**, are as follows. I have included Bhrigu with **Mizar** but he is usually thought of as the 1 of the 7 and as having the qualities of all of the 7, whereas Hewitt has **Vashishtha** which may or may not be another name for **Bhrigu**. The Vishnu Dharma claims **the star of Atri** as ruling the other stars of the bear. This list agrees with the partial list given by the Arab astronomer Al Biruni.

Dubhe, Kratu

Merak, Pulaha

Phecda, Pulastya

Megrez, Atri, the Vishnu Dharma reads that this star rules the others.

Alioth, Angiras

Mizar, (Bhrigu or Vashishtha)

Alkaid, Marichi

Traditional Hindu astronomers describe a separate precession for **the seven stars of the seven Rishis**. The precession of the equinoxes is at or about 24,000 years and the precession of the seven stars is said to be 27,000 years.

Back to the Surya Siddhanta.

9. This is that very same original textbook which the Sun God (Surya) of old promulgated: only, by reason of the revolution of the ages, there is here a difference of times.

10. Time is the destroyer of the worlds; another time has for its nature to bring to pass. This latter, according as it is gross or minute, is called by two names, real (murta) and unreal (amurta).

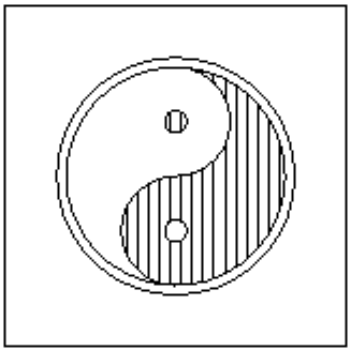
11. That which begins with respirations (prana) is called real; that which begins with atoms (truti) is called unreal. 6 respiration make a vinadi, 60 of these a nadi;

12. And 60 nadis make a day and night of respirations. Of 30 of these days of respirations is composed a month; a civil (savana) month consists of as many sunrises;

13. A lunar month, of as many lunar days (tithi); a zodiac (saura) month is determined by the entrance of the Sun God (Surya) into a sign of the zodiac: 12 months make a year. This is called a day of the Gods (sura).

Comment: Respiration, circle, and time are inseparable and directly related to each other. Brahma has the breath of the eternal, in the Timaeus of Plato the creation is said to have that one of the seven motions which is revolution in a circle, time is based on the breath of the eternal. A circle drawn on paper is seen as a spiral in time. The spiral is the oldest symbol known for time.

Time as respirations based on the breath of Brahma.



A symbol for breath from breath of the Divine.

Farther, the time occupied in pronouncing ten long syllables is said to be the time of one respiration (gurvakshara), therefore:

Putting shanthi (peace) before and after the mantra of the Buddha Avalokiteshvara we have the following ten long syllables of 'Shan thi Aum Ma ni Pad me Hum Shan thi.' To pronounce this slowly would be more or less than four seconds from one person to another but the internal search for harmony with universal design is the same for all.

The spiral of vinadi seen from two perspectives.

10 long syllables (guvakshara) x 1 breath x 6 breaths = 60

Each of 6 breaths has inhalation and exhalation for 12 units of time in 1 spiral.

From breath, to 6 breaths as a vinadi, to 60 vinadi as a nadi.

10 long syllables (guvakshara) x 1 breath x 6 breaths x 60 vinadi = 3600

Students of eastern religion should note that Ida and Pingala are called nadis. There is a relatedness of the nadi to the nadis. What that would be could only be breath regulating the awareness of time while regulating the energy flows of the meditation pattern. Meditators seek to enlongate their breathing, maximize and enlarge their energy and aura patterns, while contemplating the eternal. Inner style martial artists, such as Pau Kua, use a Hen and Ha breathing to bring the breath from breath of the Divine into harmony with the breath of the eternal. When this harmony is combined with Hen, inbreath, withdraw motion, and Ha, outbreath, extend motion, then a blow can be delivered from the center of the 7 principles of man out to the extremity of hand or foot and beyond. In this way the inner style martial artist can 'really clean the opponent's clock', so to write.

From breath, to 6 breaths as a vinadi, to 60 vinadi as a nadi, to 60 nadi as day.

10 long syllables (guvakshara) x 1 breath x 6 breaths x 60 vinadi x 60 nadi = 216,000

60 x 60 x 60 = 216,000

This basic pattern repeats on smaller and larger scales in the numbers for time. 216,000 is part of the perfect number 4,320,000 = (2 x 10) x (60 x 60 x 60)

6 breaths = 1 vinadi

60 vinadi = 1 nadi

60 nadi = 1 day of respirations

30 days of respirations = 1 month of respirations

12 months of respirations = 1 year of respirations

At 360 days there are two patterns that continue. One is out to the time of the Divine and the gods where 360 is properly understood relative to breath, circle, and time, the other is the 365 or 366 day solar year. This is what meditaters seek to do, to restore the circle of the heavens, elongate the breathing, and contemplate the eternal, that they might realize their Atmic reality while still in this life time with its solar years.

For those of us who are still working on restoring the circle of the heavens we have a year slightly longer than 360 for the time we for character development. The Taoist meditators say to distribute the difference between 365 or 366 and 360 over the four directions which are up, front, down, and back. From there a person seeks an equal distribution of energy out to all six directions while the meditation pattern slowly matures.

Spirals of years with spirals of days.

Solar time as sunrises from an old calendar system still used in India.

1 civil day = sunrise to sunrise, where the day is shorter than the night or

the day is equal to the night or

the day is longer than the night

1 solar day = midnight to midnight, or noon to noon

30 sunrises = 1 civil (savana) month

12 months = 360 days

12 months plus an intercalary month to adjust the calendar every several years.

1 solar year = winter solstice to winter solstice or is it

1 solar year = equinox to equinox?

16 phases of the moon.

Lunar time

30 lunar days (tithi) = 1 lunar month = new moon to new moon

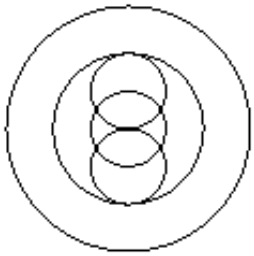
12 lunar months = 1 lunar year

The zodiac constellations, the ecliptic, and the 12 signs of 30 degrees each.

Zodiac time

The Sun God's (Surya's) eastward movement through 1 sign = 1 zodiac (saura) month

12 zodiac (saura) months = one full transit of the zodiac (Bhagana)



Symbol for the Pythagorean monad.

Divine time

1 solar year = 1 day and night of the Gods

6 x 60 solar years = 6 x 60 days and nights of the Gods = 1 Divine year

360 solar years = 1 Divine year

Apparent time with the circle with breath from breath of the Divine.

A ratio of 360 to 1.

From the above:

6 x 60 = 360 = 1 circle

This, with breath and time, seems to be the apparent connection between the solar and the Divine.

2, 216, and the powers of 10.

A set of expanded forms would be:

1 = 1

6 x 1 = 6

6 x 1 x 10 = 60

6 x 6 x 1 x 10 = 360

6 x 6 x 6 x 1 x 10 = 2,160

6 x 6 x 6 x 1 x 10 x 10 = 21,600

6 x 6 x 6 x 1 x 10 x 10 x 10 = 216,000 216,000 x 2 = 432,000

6 x 6 x 6 x 1 x 10 x 10 x 10 x 10 = 2,160,000 2,160,000 x 2 = 4,320,000 This is the perfect number.

2 times 6 cubed times 10 to the fourth, the perfect number. Timaeus speaks of the perfect number of revolutions

86,400 = 24 hours x 60 minutes x 60 seconds = seconds in a day

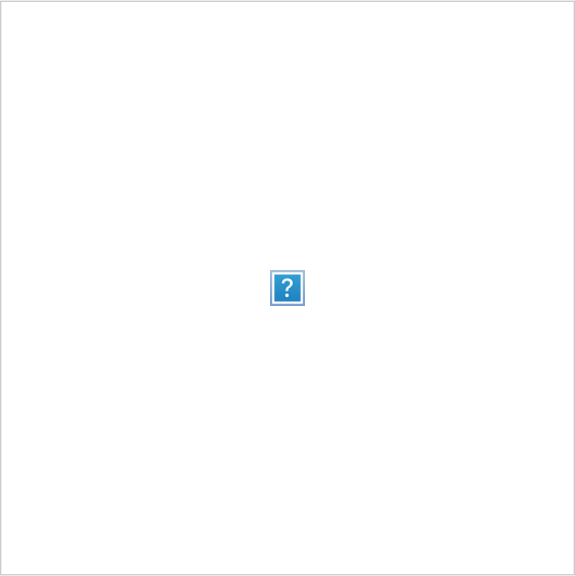
21,600 = (24 hours x 60 minutes x 60 seconds) / 4 = 4 second intervals in 24 hours

21,600 = 6 sections x 60 degrees x 60 minutes = minutes in a circle

21,600 = 6 breaths x 60 vinadi x 60 nadi = breaths in 1 day of respirations

Ideally in one day there would then be:

21,600 four second breaths while the heavens turned 21,600 minutes of the circle



21,600 four second intervals in 1 day.

21,600 four second breaths in 1 day.

21,600 minutes of arc in 1 circle.

1 respiration = 4 seconds = 1 minute of arc of the day

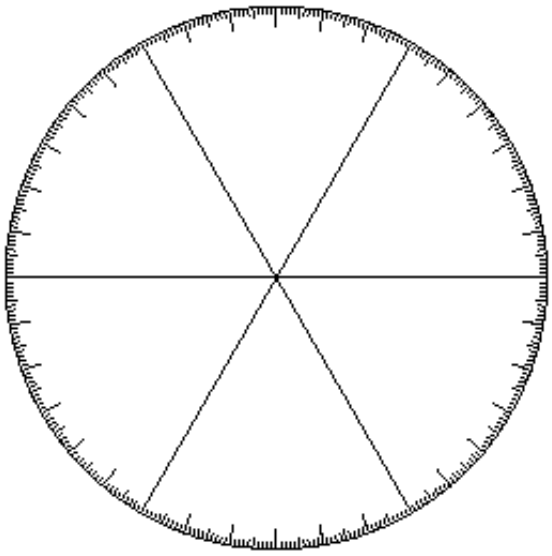
1 vinadi = 24 seconds = 6 minutes of arc of the day

1 nadi = 24 minutes = 6 degrees of arc of the day

1 day of respirations = 1 day of 24 hours = 360 degrees of arc of the day

Time both for people and planets (grahas) can be measured in respirations. I would think this is true for gods and devas and stars also.

Farther, the time occupied in pronouncing ten long syllables is said to be the time of one respiration (gurvakshara), therefore:



(10 x 6) x 60 x 60 = 60 x 60 x 60 = 216,000 long syllables possible in one day of respirations.

Putting shanthi (peace) before and after the mantra of the Buddha Avalokiteshvara we have the following ten long syllables of 'Shanthi Aum Mani Padme Hum Shanthi.' To pronounce this would be more or less than four seconds from one person to another but the internal search for harmony with universal design is the same for all.

The heart chakra is said to have time moments called thigle.

These numbers continue on into the numbers for the Yugas where is the perfect number of:

(2 x 10) x (60 x 60 x 60) = 4,320,000 years.

14. The day and night of the Gods (suras) and of the asuras are mutually opposed to one another. 6 times 60 of them are a year of the Gods, and likewise of the asuras.

Comment:

The day and night of the Gods (suras) and of the asuras are mutually opposed to one another.

The diagram looks like a picture of religion and politics balanced during the equinox.

Suras are the Gods. An a in front of a word indicates the opposite, so asuras are the opposite of the Gods. Gods are characterized by love and truth. Asuras are characterized by appetite and ambition.

The Sun God (Surya) has apparent westward motion from eastern sunrise to southern noon to western sunset each day. The Sun God (Surya) has eastern motion from Aries (Mesha) to Taurus (Vrishaba) to Gemini (Mithuna) and so on, through the 12 signs of the zodiac (bhagana) around to Aries (Mesha) again. Slightly longer is the Sun God's (Surya's) eastward motion from equinox to equinox which equals 1 solar year.

A ratio of 360 to 1.

Divine time

1 solar year = 1 day and night of the Gods

6 x 60 solar years = 6 x 60 days and nights of the Gods = 1 Divine year

360 solar years = 1 Divine year

Apparent time with the circle with breath from breath of the Divine.

15. 12,000 of these Divine years are denominated a Quadruple Age (Caturyuga); of 10,000 times 432 solar years

Comment:

1 Mahayuga = 12,000 Divine years

1 Mahayuga = 4,320,000 solar years

1 to 360 ratio

The four parts of the Quadruple Age (Caturyuga) are:

The Golden Age, (Sathya Yuga)

The Silver Age, (Treta Yuga)

The Bronze Age, (Dwapara Yuga)

The Iron Age, (Kali Yuga)

The total of these four Yuga is also called a Mahayuga, from Maha, great and yuga, age.

432 = 2 x 216 = 2 x (6 x 6 x 6)

10,000 x 432 = 4,320,000 solar years

1 Divine Year = 360 solar years

12,000 Divine years x 360 = 4,320,000 solar years

16. Is composed that Quadruple Age (caturyuga), with its dawn and twilight. The difference of the Golden (Sathya Yuga) and the other Ages (yugas), as measured by the difference in the number of the feet of Virtue in each, is as follows:

Comment:

4 footed virtue followed by 3 footed, 2 footed, and 1 footed virtue.

In the Vedanta virtue is said to walk on four feet during the Golden age (Sathya Yuga), three feet during the Silver Age (Treta Yuga), two feet during the Bronze age (Dwapara Yuga), and virtue is reduced to one foot during the Iron Age (Kali Yuga, our present yuga). Virtue then starts a new Mahayuga beginning with a new Golden Age (Sathya Yuga) where virtue once again walks on four feet.

This is the Kali Yuga, the times of one footed virtue. Honesty is said to be the predominate remaining virtue, honesty and truthfulness are a quality that can be cultivated continuously. Try reading the Bible aloud in front of people while standing on one foot, not easy.

17. The tenth part of an Age (Mahayuga), multiplied successively by 4, 3, 2, and 1, gives the length of the Golden (Sathya Yuga) and the other Ages (Yugas), in order: the sixth part of each belongs to its dawn and twilight.

Comment: Further qualities of 10 and 6 in the perfect number (2 x 10) (10 x 60) = 12,000 and (2 x 10) (60 x 60 x 60) = 4,320,000

In Divine years:

1 Mahayuga = 12,000 Divine years

1/10 x 12,000 = 1,200 Divine years

4 x 1,200 = 4,800 Divine years = The Golden Age (Sathya Yuga)

3 x 1,200 = 3,600 Divine years = The Silver Age (Treta Yuga)

2 x 1,200 = 2,400 Divine years = The Bronze Age (Dwapara Yuga)

1 x 1,200 = 1,200 Divine years = The Iron Age (Kali Yuga)

1/6 of an Age (Mahayuga) = 1/12 as dawn + 1/12 as twilight, this leaves 10/12 in between, therefore with the dawn, the duration, and the twilight we have:

The Golden Age (Sathya Yuga) = 400 + 4,000 + 400 = 4,800 Divine years

The Silver Age (Treta Yuga) = 300 + 3,000 + 300 = 3,600 Divine years

The Bronze Age (Dwapara Yuga) = 200 + 2,000 + 200 = 2,400 Divine years

The Iron Age (Kali Yuga) = 100 + 1,000 + 100 = 1,200 Divine years

In solar years:

An Age (Mahayuga) = 4,320,000 solar years

1/10 x 4,320,000 = 432,000 solar years

4 x 432,000 = 1,728,000 solar years = The Golden Age (Sathya Yuga)

3 x 432,000 = 1,296,000 solar years = The Silver Age (Treta Yuga)

2 x 432,000 = 864,000 solar years = The Bronze Age (Dwapara Yuga)

1 x 432,000 = 432,000 solar years = The Iron Age (Kali Yuga)

1/6 of an Age = 1/12 as dawn + 1/12 as twilight, this leaves 10/12 in between, therefore with the dawn, the duration and the twilight we have:

Golden Age (Sathya Yuga) = 144,000 + 1,440,000 + 144,000 = 1,728,000

Silver Age (Treta Yuga) = 108,000 + 1,080,000 + 108,000 = 1,296,000

Bronze Age (Dwapara Yuga) = 72,000 + 720,000 + 72,000 = 864,000

Iron Age (Kali Yuga) = 36,000 + 360,000 + 36,000 = 432,000

18. 1 and 70 Ages (Mahayugas) are styled here a time of a Manu (manvantara); at its end is said to be a twilight which has the number of years of a Golden Age (Sathya Yuga), and which is a deluge.

Comment: A Manu is an associate of Brahma. They work with Brahma in matters of creation and universal evolution. Manvantara means the time of a Manu.

The 14 Manvantaras of the Kalpa of Brahma.

This is the 28'th Mahayuga of the Manvantara of the Manu Vaivasvata, son of Vivaswan (Vivasvant)

Each manvantara has 70 and 1 Mahayugas and a twilight.

In Divine years:

1 Manvantara = 840,000 + 12,000 + 4800 = 856,800 Divine Years.

1 Age (Mahayuga) = 12,000

1 twilight = 4,800

(1 + 70) Ages (Mahayugas) = a time of a Manu (manvantara)

(1 + 70) Ages (Mahayugas) = 12,000 + (70 x 12,000)

(1 + 70) Ages (Mahayugas) = 12,000 + 840,000

(1 + 70) Ages (Mahayugas) = 852,000

A time of a Manu (manvantara) = 852,000 Divine years

A time of a Manu (manvantara) + a twilight = 852,000 + 4,800

A time of a Manu (manvantara) + a twilight = 856,800 Divine years

In solar years:

1 manvantara = 302,400,000 + 4,320,000 + 1,728,000 = 308,448,000 Solar Years.

1 Age (Mahayuga) = 4,320,000

1 twilight = 1,728,000

(1 + 70) Ages (Mahayugas) = a time of a Manu (manvantara)

(1 + 70) Ages (Mahayugas) = 4,320,000 + (70 x 4,320,000)

(1 + 70) Ages (Mahayugas) = 4,320,000 + 302,400,000

(1 + 70) Ages (Mahayugas) = 306,720,000

A time of a Manu (manvantara) = 306,720,000 solar years

A time of a Manu (manvantara) + a twilight = 306,720,000 + 1,728,000

A time of a Manu (manvantara) + a twilight = 308,448,000 solar years

19. In an Aeon (kalpa) are reckoned 14 such Manu with their respective twilights; at the start of the Aeon (kalpa) is a dawn, having the length of a Golden Age (Sathya Yuga).

Comment: An Aeon (kalpa) is the length of time for a universe. Therein are the times of the 14 Manu, each with their twilight, one after another, who are leading universal evolution.

The dawn of this creation, well, that's another story in the Veda of India.

In Divine Years:

1 dawn = 4,800

A time of a Manu (manvantara) = (1 + 70) Ages (Mahayugas)

1 Age (Mahayuga) = 12,000

1 twilight = 4,800

1 Aeon (kalpa) = 1 dawn + (14 Manus (1 (manvantara) + twilight))

1 Aeon (kalpa) = 1 dawn + (14 Manus ((1 + 70) Ages (Mahayugas) + twilight))

1 Aeon (kalpa) = 4,800 + (14 (12,000 + 840,000 + 4,800)

1 Aeon (kalpa) = 4,800 + (14 x 856,800)

1 Aeon (kalpa) = 4,800 + 11,995,200

1 Aeon (kalpa) = 12,000,000 Divine years

The Surya Siddhantas numbers for calculating the length of this creation in solar years.

In solar years:

1 dawn = 1,728,000

A time of a Manu (manvantara) = (1 + 70) Ages (Mahayugas)

1 Age (Mahayuga) = 4,320,000

1 twilight = 1,728,000

1 Aeon (kalpa) = 1 dawn + (14 Manus (1 (manvantara) + twilight))

1 Aeon (kalpa) = 1 dawn + (14 Manus ((1 + 70) Ages (Mahayugas) + twilight))

1 Aeon (kalpa) = 1,728,000 + (14 (4,320,000 + 302,400,000 + 1,728,000))

1 Aeon (kalpa) = 1,728,000 + (14 x 308,448,000)

1 Aeon (kalpa) = 1,728,000 + 4,318,272,000

1 Aeon (kalpa) = 4,320,000,000 solar years

20. The Aeon (kalpa), thus composed of a 1000 Ages (Mahayugas), and which brings about the destruction of all that exists, is styled a day of Brahma; his night is of the same length.

Comment:

1 Age (Mahayuga) = 12,000 Divine years

1 Aeon (kalpa) = 12,000,000 Divine years

1 Age (Mahayuga) = 4,320,000 solar years

1 Aeon (kalpa) = 4,320,000,000 solar years

1000 Mahayugas = 1 dawn + 14 Manu ((1 + 70) Mahayugas + twilight)

1000 Mahayugas = 0.4 Mahayuga + 14 Manu ((1 + 70) Mahayugas + 0.4 Mahayugas)

1000 Mahayugas = 0.4 Mahayuga + 994 Mahayugas + 5.6 Mahayugas

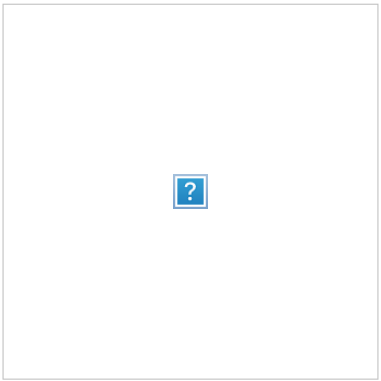
1000 Mahayugas = 994 Mahayugas + 6 Mahayugas

1000 Mahayugas = 1000 Mahayugas

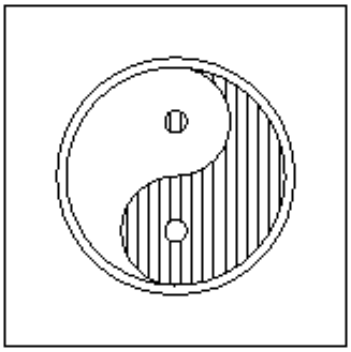
1000 Mahayugas = 1 kalpa

1000 Ages (Mahayugas) = 1 Aeon (kalpa)

The twilights at the end of the time of 13 of the 14 Manus are called either deluges or pralayas. The 14'th twilight is at the end of the day of Brahma and is said to be the major pralaya, the destruction of all that exists, the final dissolution of this universe. The pralayas at the end of each manvatara are said to be minor pralayas, that is a deluge. Instructions for the pralayas are given in the Vedanta.



The Great Symbol for Brahmaloka.



A symbol for the unity of day and night.

1 day of Brahma = 1 night of Brahma

1 day and night of Brahma = 1 day of Brahma + 1 night of Brahma

In Divine years:

1 day and night of Brahma = 12,000,000 + 12,000,000

1 day and night of Brahma = 24,000,000 Divine years

In Solar years:

1 day and night of Brahma = 4,320,000,000 + 4,320,000,000

1 day and night of Brahma = 8,640,000,000 solar years

21. His extreme age is a 100, according to this valuation of a day and a night. The half of his life is past; of the remainder, this is the first Aeon (kalpa).

Comment: Brahma's extreme age = 100 years of Brahma, from param ayuh, extreme age.

Also called para, supreme period. One half a para is called a parardha. Is dha, therefore the suffix for half?

Brahma's life = eternity

This brings to mind that half of forever has happened. The Divine memory, the eternal now, and the Divine plan. There are those in India who go on to calculate a day of Vishnu. Some would reason that a paradha of eternity, times two is eternity indeed, and as such it is only one day of Vishnu.

1 day and night of Brahma x 360 = 1 year of Brahma

1 year of Brahma x 100 = Brahma's extreme age

In Divine years:

24,000,000 x 360 = 8,640,000,000 Divine years in 1 year of Brahma

8,640,000,000 x 100 = 864,000,000,000 Divine years in Brahma's extreme age

In solar years:

8,640,000,000 x 360 = 3,110,400,000,000 solar years in 1 year of Brahma

3,110,400,000,000 x 100 = 311,040,000,000,000 solar years in Brahma's extreme age

22. And of this Aeon (kalpa), six Manu are past, with their respective twilights; and of the (current) Manu, son of Vivasvant, 27 ages are past;

Comment:

From the dawn of the creation through 6 Manus and 27 Mahayuga of Vaivasvata = 5,469,600 Divine years.

In Divine years:

1 dawn = 4,800

1 Age (Mahayuga) = 12,000

1 twilight = 4,800

A time of a Manu (manvantara) = (1 + 70) Ages (Mahayugas)

With 1 manvantara and twilight for each of 6 Manus.

dawn + 6 (manvantara) + twilight) + Vaivasvata's 27 Ages =

dawn + 6 ((1 + 70) Ages + twilight)) + Vaivasvata's 27 Ages =

4,800 + 6 (71 (12,000) + 4,800) + Vaivasvata's 27 (12,000) =

4,800 + 6 (856,800) + Vaivasvata's 324,000 =

4,800 + 5,140,800 + Vaivasvata's 324,000 = 5,469,600

dawn + 6 (manvantara) + twilight) + Vaivasvata's 27 Ages = 5,469,600

In solar years:

From the dawn of the creation through 6 Manus and 27 Mahayuga of Vaivasvata = 1,969,056,000 solar years.

1 dawn = 1,728,000

1 Age (Mahayuga) = 4,320,000

1 twilight = 1,728,000

A time of a Manu (manvantara) = (1 + 70) Ages (Mahayugas)

With 1 manvantara and twilight for each of 6 Manus.

dawn + 6 (manvantara) + twilight) + Vaivasvata's 27 Ages =

dawn + 6 ((1 + 70) Ages + twilight) + Vaivasvata's 27 Ages =

1,728,000 + 6 (71 (4,320,000 + 1,728,000) + Vaivasvata's 27 (4,320,000) =

1,728,000 + 6 (308,448,000) + Vaivasvata's 116,640,000 =

1,728,000 + 1,850,688,000 + Vaivasvata's 116,640,000 = 1,969,056,000

dawn + 6 (manvantara) + twilight) + Vaivasvata's 27 Ages = 1,969,056,000

23. Of the present, the 28'th Age (Mahayuga), this Golden Age (Sathya Yuga) is past: from this point, reckoning up the time, one should compute together the whole number.

Comment:

In Divine years:

dawn of the creation 4,800

6 Manus 5,140,800

Vaivasvata's 27 Mahayugas 324,000

sub total = 4,800 + 5,140,800 + 324,000 = 5,469,600 Divine years

The 28'th Mahayuga of Vaivasvata Manu to the present.

Sathya Yuga 4,800

yugas since the presentation in this Siddhanta:

Treta Yuga 3,600

Dwapara Yuga 2,400

Kali Yuga, Using the Hindu calendar we find, evidently, that the Iron Age (Kali Yuga) started 3102 BC.

From 3102 BC till 2003 AD

is 5,104 solar years if our calendars are correct,

at 1 Divine year to 360 solar years

5104 / 360 = 14.1777... (the 7 repeats)

0.1777... x 360 days per Divine year = 64 days of the Gods

subtotal = 4,800 + 3,600 + 2,400 + 14 and 64 days = 10,814 and 64 days

10,814 Divine years and 64 Divine days have passed of this 28'th Mahayuga of Vaivasvata.

adding the two subtotals = 5,469,600 + 10,814 and 64 = 5,480,414 Divine years and 64 Divine days

According to this calculation 5,480,414 Divine years and 64 Divine days of the Gods have of this creation have passed as of 2003 AD. Earlier we had calculated 12,000,000 Divine years for the kalpa, that is the day of Brahma, the time for this universe.

In solar years:

dawn of this creation 1,728,000

6 Manus 1,850,688,000

Vaivasvata's 27 Mahayugas 116,640,000

sub total = 1,969,056,000 solar years

The 28'th Mahayuga of Vaivasvata Manu to the present.

Sathya Yuga 1,728,000

yugas since the presentation in this Siddhanta:

Treta Yuga 1,296,000

Dwapara Yuga 864,000

Kali Yuga

From 3102 BC till 2003 AD = 5,104 solar years.

sub total = 1,728,000 + 1,296,000 + 864,000 + 5.104 = 3,893,104 solar years

3,893,104 solar years have passed of this 28'th Mahayuga of Vaivasvata.

adding the two subtotals = 1,969,056,000 + 3,893,104 = 1,972,949,104 solar years

Since the dawn of the creation, according to this calculation 1,972,949,104 solar years have passed up till 2003 AD. Earlier we had calculated 4,320,000,000 solar years for the kalpa.

Checking our answers:

Divine years passed / Divine years in kalpa = solar years passed / solar years in kalpa

5,480,414.17... (repeats 7) / 12,000,000 = 1,972,949,104 / 4,320,000,000

0.4567011814...(repeats 814) = 0.4567011814...(repeats 814)

This also reads that a little more than 45 % of the time (kalpa) for this universe has passed.

Another check:

solar years / 360 = Divine years

1,972,949,104 / 360 = 5,480,414.17...

From this point, reckoning up the time one should compute together the whole number. Therefore: Brahma's extreme age (param ayuh) is a 100, according to this valuation of a day and a night, the half of His life (parardha) is past; of the remainder, this is the first Aeon (kalpa). And of this Aeon (kalpa), 6 Manu are past, with their respective twilights; and of the Manu Vaivasvata, 27 Ages (Mahayuga) are past; of the present, the 28'th, Age (Mahayuga), this Golden Age (Sathya Yuga) is past. Since then the Silver Age (Treta Yuga) is past, the Bronze Age (Dwapara Yuga) is past, and that part of the Iron Age (Kali Yuga), from 3102 BC, if I understand the Hindu calendar correctly, until 2003 AD, 14 Divine years and 64 days of the gods, which are 5,104 solar years, are past. For a total of 5,480,414 Divine years and 64 days of the gods, which are 1,972,949,104 solar years, of this Aeon (kalpa) are past.

This calculation is called the sum of years, it can be extended into the sum of days. This is a standard calculation for esoteric astronomers both in Hinduism and the Hebrew kabbala. Nostradamus presents a sum of days in his Epistle to Henry II.

24. 100 times 474 Divine years passed while the all wise was employed in creating the animate and inanimate creation, plants, stars (bha), gods (suras), asuras, and the rest.

Comment: In Divine years:

100 x 474 = 47,400 Divine years of creating the creation

The Genesis question is brought to mind, is that 6 equal days or units for creating? And if so then:

47,400 / 6 = 7900

6 equal units of 7900 Divine years for each during the time of creation.

Numerically the following ratios are evident:

1 / 80 = 600 / 48,000

79 / 80 = 47,400 / 48,000

Dawn 4,800, creating starts at dawn

1'st Age (Mahayuga) 12,000

2'nd Age (Mahayuga) 12,000

3'rd Age (Mahayuga) 12,000

4'th Age (Mahayuga)

Golden Age (Sathya Yuga) 4800

Silver Age (Treta Yuga) 3600

Bronze Age (Dwapara Yuga) 2400

1/2 Iron Age (Kali Yuga) 600

4800 + 3600 + 2400 + 600 = 47,400

Total 47,400 Divine years creating

47,400 Divine years ended at 1/2 of the Iron Age (Kali Yuga) of the 4'th Mahayuga, for 3.95 Mahayugas.

3.95 / 4 = 79 / 80

47,400 / 48,000 = 79 / 80

25. The planets, moving westward with exceeding velocity, but constantly beaten by the stars (nakshatra), fall behind, at a rate precisely equal, proceeding each in its own path.

Comment: This is similar to humans watching a western sunset while thinking about the eastern Buddha.

The planets receive corrective beatings by the stars.

The 27 and 1 nakshatra, lunar constellations.

Nakshatra refers to a group of 27 and 1 constellations that form the lunar hoop, called the moon mansions of Chandra, the lunar goddess. They accompany the zodiac signs (rasi) of 12 constellations. Stars in general are called bha. The demigods are with their spheres, hence one can feel that the Sun God (Surya) is indeed with the sun, and that Mercury (Budha) is with his sphere called Mercury (Budha), and that Bhumi the earth mother goddess is with her sphere called the earth. But this does not seem to be so with the moon. Evidently Chandra, the lunar goddess, has graduated and left behind a vacant moon. Chandra is said, therefore, to dwell in the mind and is considered omnipresent and yet mind is said to be the moon. Soma is the moon god.

27 of 28 nakshatra form a continuous hoop with the 22'nd being the uniting constellation, for 27 and 1. With students this lunar hoop of constellations is for the development of the mind and is the single most needed addition to western astrology. The Arab astronomers have a similar hoop of lunar manzils, and the Chinese have a similar hoop of lunar hsiu.

The path of the Sun is the ecliptic through the zodiac with several nearby constellations shown.

The 27 nakshatra are in or near the ecliptic as a lunar hoop to accompany the zodiac hoop.

Abhijit in Lyra is the unifying nakshatra for the whole hoop.

The Nakshatra constellations, their location to the zodiac constellations, and a brief description are:

1. **Ashvini,** two stars (bha bha) called the twin horsemen, at the start of Aries (Mesha), balanced by the twins Castor and Pollux at the end of Gemini.

2. **Bharini,** its ruling deity is Yama, the Divine as his own counterpart, the god of death and judgment. 3 stars (bha) toward the east end of Aries (Mesha) thought of as a pendulum or a yoni.

3. **Krittika,** 7 stars (bha) of the Pleiades above Taurus (Vrishaba). Called Agneya, the female form of Agni, fire, the ruling deity. Thought of as a little knife, or a cutting quality. In the Greek the names of the 7 sisters are Atlas, Alcyone, Merope, Electra, Maia, Asterope, Taygete. Merope is said to be the junction star (yogatara) of the nakshatra. Pleione and Celaeono are nearby. Job 38: 31 "Can you bind the beautiful Pleiades."

4. **Rohini,** 'ruddy.' The ruling deity is Prajapati, the lord of all created beings. Said to be the favorite nakshatra. It is Aldebaron (Agni) and the Hyades, the bright red star (yogatara), said to be the dominant star (bha) of the eastern quadrant, also called the 'bull's eye,' the lower tip of the 5 stars (bha) in the V of Taurus (Vrishaba). The Hyades are a star cluster like the Pleiades and can easily be seen with binoculars.

5. **Mrigasiras,** 'antelopes head.' Its ruling divinity is Soma, the god of the moon. It is the head of Orion, below Taurus (Vrishaba), Job 38: 31 "Can you bind the beautiful Pleiades. Can you loose the cords of Orion."

6. **Ardra,** 'the gem.' Its ruling divinity is Rudra, the storm god. It is the armpit of Orion, at Betelgeuse below Taurus (Vrishaba), said to give strength as well as wisdom.

7. **Punarvasu,** punar, again and vasu, good. Its ruling divinity is Aditi, the mother. The twins, Castor, the upper star (bha), and Pollux, the lower star (yogatara), in Gemini (Mithuna),

8. **Pushya,** push, nourish or thrive. Its ruling divinity is Brihaspati, the priest and teacher (guru). It is the middle of Cancer (Kataka), three stars (bha) quite on and near the ecliptic.

9. **Aslesha,** associated with sarpas and nagas, serpents and snakes. It is the loop in the head of Hydra below Cancer (Kataka). Hydra is like a watery border under Virgo (Kanya), Leo (Simha), and the water sign Cancer (Kakata).

10. **Magha,** 'mighty,' 'mighty houses' The fathers, the ancestors, are the regents. The head of Leo (Simha), has the northern dominant, the star (yogatara) Regulus, said to give courage as well as wisdom.

11. **Purva Phalguni,** 'place of sitting.' The first part of the triangle in Leo (Simha), Purva means former or preceding or southern.

12. **Uttara Phalguni,** 'place of sitting.' The last part of the triangle in Leo (Simha), Uttara means latter or following or northern.

13. **Hasta,** 'hand.' Its ruling deity is the Sun God (Surya) also called Savitar. Below Virgo (Kanya) in Corvus, at the tail end of Hydra. Corvus is called the raven of Apollo in the Greek.

14. **Chithra,** 'brilliant.' Its ruler is Tvashtar, the shaper and fashioner. It is Spica (Chithra) in Virgo (Kanya). Spica (Chithra) is the star (yogatara) of the start of Libra (Thula) and was 180 degrees from the start of Aries (Mesha) during the times of this Siddhanta. Now its longitude is 183 degrees 49'. Spica in the Greek, a favorite star, was thought of as a virgin with a sheaf of wheat.

15. **Svati,** Arcturus above Virgo (Kanya) in Bootes. Follow the arc of the handle of the Big Dipper to Arcturus. Called the great one by this Siddhanta. Called the shepard's star throughout the middle east. Job 38: 31 "Can you bind the beautiful Pleiades? Can you loose the bonds of Orion? Whose to lead the sons of Arcturus? And whose to lead Mazoroth (Ursa Major) out in due season?"

16. **Vrischika,** 'having spreading branches.' Its ruling divinity is Indragni, the god of eastern fire. It is in Libra (Thula), the first part of the scales.

17. **Anuradha,** 'success.' Its ruler is Mitra, friend. Three stars (bha) in the head of Scorpio (Vrischika)

18. **Jyeshtha,** the oldest, ruddy like Rohini. Its ruler is Indra, the god of the east. It is at Antares, the bright red star (yogatara) said to be the dominant in the western quadrant, in mid Scorpio (Vrischika)

19. **Mula,** 'root.' Its ruler seems to be Varuna, the god of the west. 9 stars (bha) in a long curve, the tail of Scorpio (Vrischika)

20. **Purva Ashadha,** 'ability.' Its ruling divinity is Apas, the god of the waters. It is the front of Sagitarrius (Dhanus)

21. **Uttara Ashadha,** its ruling divinities are suras, the gods. It is the back of Sagitarrius (Dhanus)

Lyra

22. **Abhijit,** ‘conquering.’ It ruling divinity is Brahma. It is called the star (yogatara) of Krishna. It is Vega (Abhijit) in Lyra high above Sagitarrius (Dhanus), the star (yogatara) of the musicians and the devas. It is that nakshatra which is above the circle of 27 nakshatras.

23. **Shravana,** 'hearing.' Its ruler is Vishnu. Associated both with the three steps of Vishnu striding through the heavens and the little trident on the long staff that the Brahmins carry. It is 3 stars (bha) at Altair in Aquila, the eagle above Capricorn (Makara)

24. **Shravishtha,** its ruling deities are the Vasus, those who are shining and brilliant. It is 4 lovely stars (bha) at Delphinus, the dolphin, above Aquarius (Kumbha). It resembles the Pleiades (Krittika), like a little kite. One of the most beautiful of constellations. The Greeks admired it also. The Greeks had Delphi and the Delphic oracle.

25. **Shatabhishaj,** a hundred stars (bha), 'the hundred physicians.' Its ruler is Varuna, the god of the west. It is near or on the ecliptic in Aquarius (Kumbha)

26. **Purva Bhadrapada,** 'beautiful foot.' It is the first part of the square (catur) of Pegasus above Pisces (Mina)

27. **Uttara Bhadrapada,** 'beautiful foot.' It is the second part of the square (catur) of Pegasus above Pisces (Mina)

28. **Revati,** 'daughter of the king (Raja) who shines on heaven (Suvaha and Bhuva) and earth (Bhumi).' Its ruler is Pushan, the spiritually wealthy. It is in Pisces (Mina) where the angled string crosses the ecliptic, at Zeta Pisces (Revati). 10 minutes east of this star (yogatara) is the end of Pisces (Mina) and the start of Aries (Mesha). Revati as a constellation has 32 stars (bha) thought of as a drum.

Learning the nakshatra will increase a person's mental abilities. Taking a flashlight with a red cloth taped over it to minimize the glare, and a star map, and one's new found mental abilities out to the field to look at the stars, a person can then learn to identify all the zodiac and nakshatra by sight. So doing establishes that communion with the heavens that leads a person out of the darkness and into the light. At length a type of stellar clairvoyance will develop in the student.

26. Hence they have an eastward motion. From the number of their revolutions is derived their daily motion, which is different according to the size of their orbits; in proportion to this daily motion they pass through the stars (nakshatra).

Comment: The planets (grahas) have apparent westward motion during the course of a day or night and eastward motion through the zodiac (bhagana) during the course of a year.

'Hence the planets have an eastward motion' also seems to refer to the planets having been beaten by the stars, which convinces them, evidently to take up a dutiful eastward motion.

India's astronomers have described the planets (grahas). Two of these, Varahamihira who wrote the Brihat Jataka and Mantreswara who wrote the Philadepika, have been quoted by Ronnie Gale Dreyer in her book, Vedic Astronomy. From that book:

Varahamihira on the Sun God (Surya): The Sun has somewhat yellow eyes, is of the height of the length of the two arms stretched out, of bilious nature and with very little hair on his head.

Mantraswara on the Sun God (Surya): The Sun is of bilious temperament, Lord of bones, limited quantity of hair, dark red form, reddish brown eyes, red, square built body, valiant, wrathful broad shoulders. He is a Shiva worshipper, doctor, minister, king, performer of sacrifice. It has to do with matters concerning the self, copper, gold, father, auspiciousness, happiness, dignity, power, glory, influence, health, vigor, fortune, and the God Shiva.

The Sun God (Surya) is said to rule Leo (Simha).

Varahamihira on Chandra, the lunar goddess: The Moon has a thin and round body, is of an exceedingly windy and phlegmatic nature, is learned and has a soft voice and beautiful eyes...

Mantraswara on Chandra, the lunar goddess: The Moon has a huge body, is young and old, lean and white, has fine lovely eyes, black and thin hair, governs blood, is soft in speech, wears white, is yellowish in color, has wind and phlegm in his composition and is mild in temperament. It has to do with matters concerning mother, mental tranquility, fruits, tenderness, flowers, fame, pearls, milk, sweet substances, women, bodily health, beauty, heart, understanding, and affluence. The Moon is female and is concerned with the sea.

Chandra the lunar goddess is said to rule Cancer (Kakata).

Varahamihira on Mars (Mangal): Mars has sharp and cruel eyes and a young body, is generous, of bilious nature, of unsteady mind, and has a narrow middle.

Mantraswara on Mars (Mangal): Mars has a slender waist, curled and shining hair, fierce eyes, is bilious and cruel in nature, red, wrathful but generous. It rules matters pertaining to strength, products from the earth, brothers, war, courage, fire, enemies, attachment to females, mental dignity. It has to do with courage, disease, younger brothers, lands, foes, blood, paternal relations. He is a crook, and an arms bearer, a goldsmith, and a thief.

Mars (Mangal) is said to rule Aries (Mesha) and Scorpio (Vrischika).

Varahamihira on Mercury (Budha): Mercury has an impediment in his speech, is fond of joke, and is of a bilious, windy and phlegmatic nature...

Mantraswara on Mercury (Budha): Mercury is green, full of nerves, pleasant in speech, red and broad eyes, and is fond of fun. It has to do with matters pertaining to learning, eloquence, skill in fine arts, dexterity in speech, maternal uncle, aptness for acquiring knowledge, cleverness and the mechanical arts. It has to do with relatives, discrimination, friends, speech and action. He is a cowhand, learned man, artisan, accountant.

Mercury (Budha) is said to rule Gemini (Mithuna) and Virgo (Kanya).

Varahamihira on Jupiter (Guru): Jupiter has a big body, yellow hair and eyes, high intellectual powers, and a phlegmatic nature...

Mantraswara on Jupiter (Guru): Jupiter has body of yellowish hue and eyes and hair are brown, fat and elevated chest, big body, phlegmatic, intelligent, and is after wealth. It has to do with knowledge, good qualities, teaching, prosperity, reverence to Gods, wisdom, and conquering of the senses. He is an astrologer, preceptor, minister, important personage, and Brahmin. It is genius, wealth, physical development, sons and knowledge. The sense it rules is hearing.

Jupiter (Guru) is said to rule Pisces (Mina) and Sagittarius (Dhanus).

Varahamihira on Venus (Sukra): Venus leads a comfortable life, has a beautiful body, fair eyes, a windy and phlegmatic nature, and black curling hairs...

Mantraswara on Venus (Sukra): Venus wears multi colors, black curled hair, limbs and body are huge, wind and phlegm, green and lovely and broad eyes, and he treasures his virility. It has to do with matters pertaining to clothes, song and dance and music, wife, happiness, flowers, poetry, charming speech, and marriage. Venus is a musician, wealthy man, sensualist, merchant, dancer, weaver, peacock. It has to do with wives, vehicles, ornaments, love affairs, pleasure.

Venus (Sukra) is said to rule Taurus (Vrishaba) and Libra (Thula).

Varahamihira on Saturn (Shani): Saturn is lazy, has eyes of gold color, a thin and tall body, large teeth, stiff hair, and is of a windy nature.

Mantraswara on Saturn (Shani): Saturn is dark, lame, windy in composition, deep eyes with lean tall body full of arteries and veins, idle and calumniating, governs muscle, cruel and has no pity, dull headed, large nails, teeth, stiff hair and limbs, dirty and of a slow disposition, fierce and black. It has to do with longevity, death, fear, misery, sickness, misfortune, servitude, captivity, and poverty. It defines the period of a person's life, livelihood, cause of death, adversity and servants. Saturn is an oil monger, servant, vile person, blacksmith.

Saturn (Shani) is said to rule Aquarius (Kumbha) and Capricorn (Makara).

The Sun God (Surya) rules Leo (Simha)

The moon (Chandra) rules Cancer (Kakata)

Mercury (Budha) rules Virgo (Kanya) Gemini (Mithuna)

Venus (Sukra) rules Libra (Thula) Taurus (Vrishaba)

Mars (Mangal) rules Scorpio (Vrischika) Aries (Mesha)

Jupiter (Guru) rules Sagittarius (Dhanus) Pisces (Mina)

Saturn (Shani) rules Capricorn (Makara) Aquarius (Kumbha)

So Jupiter is said to have studied the Veda and become a Brahmin. As such he is called Guru, the teacher. Most all experienced astronomers take up the study of the 60 cycle of Jupiter, both in India and Greece. As a person comes into agreement with Guru it is possible to feel the larger time cycles of 60, 3600, and 216,000 years. Guru is said to be an astrologer, so then we have planets that study astrology? Evidently.

27. One which moves swiftly passes through them in a short time; one which moves slowly, in a long time. By their movement, one full transit is accounted complete at the end of the constellation Revati (nakshatra).

Comment: One full transit as bhagana means 'enjoyed the whole troop of stars (bha).' One full transit is the apparent motion, when viewed from the earth (Bhumi) from the start of Aries (Mesha), around eastward, to the start of Aries (Mesha) again. Zeta Pisces (Revati) is on the angled string, of the two strings of Pisces (Mina), quite near the ecliptic. The constellation (nakshatra) and the star (bha) are both called Revati. The constellation Revati (nakshatra) starts within Pisces (Mina) at 346 degrees 40' and stops at 360 degrees. The star Zeta Pisces (Revati) is at 359 degrees 50'. Indian astronomers say the start of Aries (Mesha) is 10' east of Zeta Pisces (Revati). Others say Aries (Mesha) ends 15 degrees west of Aldebaran (Agni); this latter is the more accepted arrangement.

Tradition has Aldebaran centered in Taurus. This puts the start of Ares 23 degrees 51' east of the equinox.

This is the ayanamsa, the adjustment for the equinox, that eastern astrology uses.

28. 60 seconds (vikala) make a minute (kala); 60 of these, a degree (bhaga); of 30 of the latter is composed a sign (rasi); 12 of these are a zodiac (bhagana).

Comment:

Zodiac constellations with 12 zodiac signs of 30 degrees each, 27 and 1 nakshatra, and the ecliptic.

60 seconds (vikala) = 1 minute (kala)

60 minutes (kala) = 1 degree (bhaga)

30 degrees (bhaga) = 1 sign (rasi)

12 signs (rasi) = 1 zodiac (bhagana)

bha, star

bhaga, degree

bhagana, one full transit of the zodiac

kala, time

vikala, 60 seconds of arc

kala, 1 minute of arc

rasi, sign of the zodiac

dinarasi, sum of days

A famous learned paper on astrology from India is Parasara's Hora Sastra, sometimes spelled Hora Shastra. An hora is one of two parts of a sign (rasi). Hora Shastra is then the learned paper on the two parts of each sign (rasi).

29. In an Age (yuga), the revolutions of the Sun (Surya), Mercury (Budha), and Venus (Sukra), and of the revolutions of the node of the conjunctions (sighra) of Mars (Mangal), Saturn (Shani), and Jupiter (Guru), moving eastward, are 4,320,000.

Comment: We note that the Surya Siddhanta is a geocentric astronomy. It would seem that the Sun God (Surya), with Venus (Sukra), and Mercury (Budha) are going around the earth (Bhumi) together and therefore have the same number of revolutions. While going around the Sun God (Surya) Venus (Sukra) and Mercury (Budha) are first going toward the earth (Bhumi), then to near the earth (Bhumi) where is inferior conjunction, then around the Sun God (Surya) to the far side where is superior conjunction, then back around toward the earth (Bhumi) again. Meanwhile our reference, the earth (Bhumi), is holding still. Mars (Mangal), Jupiter (Guru), and Saturn (Shani) go around both the earth (Bhumi) and the Sun God (Surya), each having a superior conjunction, that is when a planet (graha) is on the far side of the Sun God (Surya). With Mars (Mangal), Saturn (Shani), and Jupiter (Guru) this happens once a solar year. Therefore their conjunctions equal that of the Sun God's (Surya's) revolutions.

A geocentric cosmos (Brahma) is diagrammed by way of epicycles; an orbit, the planet's (graha's), accompanying an orbit, the Sun God's (Surya's). An orbit, then is different than a full transit of the zodiac (bhagana). A planet (graha) in an epicycle, in a geocentric system (Brahma), when viewed from the earth (Bhumi), makes a full transit of the zodiac (bhagana) from the start of Aries (Mesha) eastward around the zodiac to the start of Aries (Mesha) again. The orbit of a planet (graha) could be thought of as an ellipse that moves with the ellipse of the Sun God (Surya) that moves around the earth (Bhumi) within the circle of the heavens, a geocentric cosmos (Brahma) with epicycles.

Each of the elliptical orbits has three nodes that are thought of as stationed in the circle of the heavens. These are the node of the conjunction, the node of the apsis, and the node of the declination. These three nodes revolve about the circle of the heavens and the planet's (graha's) elliptical orbit is contained within this set of limits. The Siddhanta goes on to give the revolutions of each.

First some definitions:

Longitude. Great circle through the north and south pole for measuring distance east or west from a prime meridian. The southern meridian can be measured from in a like manner, around on the horizontal both east and west. Longitude can be measured around the ecliptic from the start of Aries (Mesha), usually measured as eastward longitude.

Declination. The planets (grahas) are seen as having declination, that is, below, on, or above the ecliptic. Declination is usually measured as north, above, the horizon. The planets can be seen as having declination from the ecliptic also, that is, below, on, or above the ecliptic.

Node of the conjunction. A planet is in superior conjunction when it is on the far side of the Sun God (Surya) when viewed from the earth (Bhumi). The node of the conjunction is stationed beyond the planet in the circle of the heavens. The node of the conjunction revolves eastward around the circle of the heavens. When the node of the conjunction has revolved half way around the planet is nearest the earth in inferior conjunction. The node of the

conjunction revolves half way around again and the planet moves to superior conjunction once again.

Node of the apsis. With apogee, aphelion, perigee, perihelion. Apsis is the long side of an elliptical orbit where is the apex called the apogee or the aphelion. When orbiting the earth (Bhumi) it is called the apogee (gee as in geography), when orbiting the Sun God (Surya) it is called the aphelion (helion as in helios). Conversely, the short side of the long axis of an elliptical orbit has an apex called the perigee when orbiting the earth (Bhumi), called perihelion when orbiting the Sun God (Surya). Out from apogee or aphelion of a planet is the node of the apsis stationed in the circle of the heavens. As the node of the apsis revolves eastward around the circle of the heavens the apogee of the orbit moves about with it.

Node of the declination. The planet's orbits are tilted slightly to the ecliptic. The ecliptic is one full transit of the Sun God (Surya) in front of the zodiac as seen from the earth (Bhumi). The planets are seen as having declination, that is, below, on, or above the ecliptic. When a planet's orbit goes from below the ecliptic, to above the ecliptic there, in the circle of the heavens, is the ascending node. This is called the planet's node of the declination. When a planet goes from above the ecliptic to below the ecliptic there is the descending node. The node of the declination revolves retrograde, westward. The Sun God (Surya) remains on his ecliptic, therefore he has no observable node

Later in the document node of the conjunction, node of the apsis, and node of the declination are further defined as structures of a higher heaven.

More definitions:

Ecliptic. The path of the ecliptic is one full transit of the Sun God (Surya) in front of the zodiac as seen from the earth (Bhumi).

Orbit. The path of the Sun God (Surya) could be called an orbit, but it is rather like the journey of a charioteer, as described in the Srimad Bhagavada as well as by Plato. The orbit of the moon (Chandra) is her journey around the earth, and for Venus (Sukra) and Mercury (Budha) it is their journey around the Sun God (Surya). The outer planets journey around both the Sun God (Surya) and the earth. The significance in metaphysics is in acknowledging the earth to be the earth mother Goddess Bhumi. The Goddess is as old as the God. The Sun God (Surya) and the planets (grahas) go about their celestial courses within the design provided for them. Conjunction on one side and near the earth on the other is much more basic to their origins than just plain physics of orbits. What then are their origins? Well that is another question, back to orbits. The outer planets journey around both the Sun God (Surya) and the earth (Bhumi). I started with orbit, but it still seems more like a journey eastward.

One full transit. A planet's journey, viewed from the earth, from the start of Aries (Mesha)

once eastward around the zodiac to the start of Aries again.

Revolution. Defining revolution has proven the most difficult item so far. All of the calculations are based on a proper understanding of universal design. From sloka 29, one revolution is once around the earth (Bhumi). For the Sun God (Surya), the moon (Chandra), and the planets, the different motions are eastward yearly and the apparent motions are westward daily. For the stars the eastward motion results in the precession of the equinoxes and the apparent daily motion is westward.

A revolution is either once around the earth (Bhumi) or one full transit of the zodiac (bhagana), one way or the other it would change the calculations. I'll return to the question of revolutions in verse 34.

If I understand this correctly and assuming that the earth (Bhumi) is holding still, then:

For the Sun God (Surya) one revolution is his eastward journey once from equinox to equinox. One revolution is longer than one full transit due to the precession of the equinoxes. One westward revolution is from above a southern longitude of the earth once around to above the same longitude of the earth again, thought of as midnight to midnight or noon to noon. Sunrise to sunrise varies throughout the seasons.

For the moon (Chandra) one eastward revolution is her eastward motion once around the earth. One revolution of the moon is shorter than new moon to new moon which, in turn, is longer than one full transit of the moon. For the moon (Chandra) one revolution westward is from above a southern longitude of the earth around to above that same longitude again.

For the superior planets, Mars (Mangal), Jupiter (Guru), and Saturn (Shani) one eastward revolution is the eastward journey once around both the earth (Bhumi) and the Sun God (Surya). One revolution is longer than one full transit. For these one revolution westward is from above a southern longitude of the earth around to above that same longitude again.

With Mercury (Budha) and Venus (Sukra) one eastward revolution is once eastward around the earth (Bhumi) and it is stated to be the same number in one Age (Mahayuga) as the number of the eastward revolutions of the Sun God (Surya) in an Age (Mahayuga). For these two the revolution of the node of conjunction is also counted. One eastward revolution of the node of the conjunction is the from the planet's alignment on the far side of the Sun God (Surya), as viewed from the earth (Bhumi), to moving eastward around once again to alignment on the far side of the Sun God (Surya) again. For these two, one eastward revolution of the node of the conjunction is shorter than one orbit and one orbit is shorter than one full transit. That is hard to visualize. Starting from the superior conjunction Mercury would journey eastward through the zodiac until the tangent of his orbit is pointing directly at the earth, from there he would have apparent retrograde motion until the tangent of his orbit is pointing directly away from the earth, from there he would have eastward motion through the

zodiac until he is at superior conjunction again, from there he would continue eastward until he completed one orbit of the Sun God Surya, from there he would continue eastward and retrograde alternately until he completed one full transit of the zodiac (bhagana). For these, also, one revolution westward is from above a southern longitude of the earth around to above that same longitude again.

One eastward revolution of the stars is one eastward journey of the stars around the earth. A visible result of this is the precession of the equinox. For the stars, also, one revolution westward is from above a southern longitude of the earth around to above that same longitude again.

30. Of the moon's revolutions, 57,753,336; of Mars (Mangal), revolutions, 2,296,832;

Comment: A skilled mathematician might be able to work out the relation between revolutions and musical notes.

31. Of Mercury's (Budha's) revolutions of the node of the conjunction (sighra), 17,937,060; of Jupiter (Guru), revolutions, 364,220;

32. Of Venus's (Sukra's) revolutions of the node of the conjunction (sighra), 7,022,376; of Saturn (Shani), revolutions, 146,568;

33. Of the moon's (Chandra's) revolutions of the node of the apsis (ucca), in an Age (Mahayuga), 488,203; of the moon's (Chandra's) node of declination (pata), in the contrary direction, 232,238;

34. Of the stars (nakshatra), revolutions, 1,582,237,828. The number of risings of the stars (nakshatra), diminished by the number of the revolutions of each planet respectively, gives the number of risings of the planets in an Age (Mahayuga).

Comment: It seems it would then be westward revolutions of the stars with an eastward revolution of the planets. Revolution then must be once around the earth both for the stars westward motion and the Sun God's eastward motion. One revolution of the sun would then equal his journey from equinox to equinox and one revolution of the stars would equal their movement from over a line of longitude of the earth around westward and back over the same longitude again. This only if the earth is holding still. So assuming that the earth is holding still:

Risings of the stars (nakshatra) - revolutions of each planet respectively = risings of the planets in an Age (Mahayuga).

1,582,237,828 = star risings

1,582,237,828 - 4,320,000 = 1,577,917,828 of the Sun God's (Surya's) risings

1,582,237,828 - 57,753,336 = 1,524,484,492 of the moon's risings

1,582,237,828 - 2,296,832 = 1,579,940,996 of Mars' (Mangal's) risings

1,582,237,828 - 17,937,060 = 1,564,300,768 of Mercury's (Budha's) risings

1,582,237,828 - 364,220 = 1,581,873,608 of Jupiter's (Guru's) risings

1,582,237,828 - 7,022,376 = 1,575,215,452 of Venus' (Sukra's) risings

1,582,237,828 - 146,568 = 1,582,091,260 of Saturn's (Shani's) risings

1,582,237,828 - 488,203 = 1,581,749,625 of the moon's node of apsis risings

1,582,237,828 - (- 232,238) = 1,582,470,066 of the moon's node of declination risings

1,577,917,828 of the Sun God's (Surya's) risings in a Mahayuga /

4,320,000 revolutions of the Sun God (Surya) in a Mahayuga =

civil days per solar year = solar days per solar year

1,577,917,828 / 4,320,000 = 365. 258756481... civil days per solar year

Risings of the Sun God (Surya) per Age (Mahayuga) = civil days per Age (Mahayuga)

Civil days per Age (Mahayuga) / revolutions of the moon = civil days per revolution

1,577,917,828 / 57,753,336 = 27.321674... civil days per revolution of the moon

= 27.321674... solar days per revolution of the moon

In a like manner for Mars and the others:

1,577,917,828 / 2,296,832 = 686.9974... solar days per revolution of Mars (Mangal)

1,577,917,828 /17,937,060 = 87.9697... solar days per revolution of the node of

the conjunction for Mercury (Budha)

1,577,917,828 / 364,220 = 4332.3206... solar days per revolution of Jupiter (Guru)

1,577,917,828 / 7,022,376 = 224.69856..solar days per revolution of the node of

conjunction for Venus (Sukra)

1,577,917,828 / 146,568 = 10765.773... solar days per revolution of Saturn (Shani)

1,577,917,828 / 488,203 = 3232.09367...solar days per revolution of the moon's

node of the apsis

1,577,917,828 / 232,238 = 6794.3998... solar days per revolution of the moon's

node of the declination

Per revolution, once around the earth, is longer than per orbit.

35. The number of lunar months is the difference between the number of revolutions of the Sun God (Surya) and of the moon (Chandra). If from it the number of civil months be subtracted, the remainder is the number of intercalary months.

Comment:

Number of revolutions of the moon (Chandra) = 57,753,336

Number of revolutions of the Sun God (Surya) = 4,320,000

Revolutions of the moon (Chandra) - Revolutions of the Sun God (Surya) = lunar months per Mahayuga

57,753,336 - 4,320,000 = 53,433,336 lunar months per Mahayuga

12 x solar years in a Mahayuga = civil months in a Mahayuga

12 x 4,320,000 = 51,840,000 civil months in a Mahayuga

Lunar months per Mahayuga - civil months in a Mahayuga = intercalary months

53,433,336 - 51,840,000 = 1,593,336 intercalary months

Reversing the calculation we find that 1,593,336 intercalary months are added, 1 about every five years, to 51,840,000 solar months to make the lunar calendar coincide with the solar calendar. The result is sometimes called the lunar solar calendar.

Time is Divine in its origin and expression. In India the higher worlds of the devas are said to keep time in reference to the moon. Such is time of the lunar moon mansions of Chandra.

The world keeps time in reference to the sunrises. The calendar of India is first Divine in its origin, then lunar, with lunar months and days, then solar with months and civil days. The lunar and the solar are made to coincide with intercalary months and omitted days.

If I understand this correctly the sequence is: With 12 months of 30 days, followed by 12 months of 30 days, the beginning day of the second set doesn't coincide with the solstice or the equinox of the second set. It is 5 days short. After two years it is 10 days short; three years, 15; four years, the leap year, 21; and after 5 years it is 26 days short. A 13'th month of 30 days is then added. The process repeats and an intercalary month is added according to a schedule. I believe every so many decades even another type of intercalary month is needed.

Since there are more lunar days, which are shorter, than civil days which are longer, then at times part of two lunar days will happen during the same civil day. When this happens only one of the two lunar days, the one wherein is sunrise, are said to coincide with the civil day. By omitting one of two lunar days the lunar calendar is made to coincide with the solar.

36. Take the civil days from the lunar, the remainder is the number of omitted lunar days (tithikshaya). From rising to rising of the Sun (Surya) are reckoned civil days;

Comment:

Lunar days - Civil days = omitted lunar days

37. Of these there are, in an Age (Mahayuga), 1,577,917,828 (civil days); of lunar days, 1,603,000,080;

Comment:

1 Mahayuga = 4,320,000 solar years

53,433,336 lunar months x 30 = 1,603,000,080 lunar days per Mahayuga

1,582,237,828 revolutions of stars - 4,320,000 revolutions = 1,577,917,828 civil days

Lunar days - civil days = omitted lunar days

1,603,000,080 - 1,577,917,828 = 25,082,252 omitted lunar days

Civil days / years in Mahayuga = civil days per year

1,577,917,828 / 4,320,000 = 365.258756481... civil days per year

Civil days / lunar months = number of civil days from new moon to new moon

1,577,917,828 / 53,433,336 = 29.53... civil days from new moon to new moon

38. Of intercalary months, 1,593,336; of omitted lunar days, 25,082,252;

39. Of solar months, 51,840,000. The number of risings of the stars (nakshatra) diminished by that of the revolutions of the Sun God (Surya) gives the number of civil days.

Comment:

1 Mahayuga = 4,320,000 solar years

Revolutions of the Sun God (Surya) x 12 solar months = solar months per Mahayuga

4,320,000 x 12 = 51,840,000 solar months per Mahayuga

Risings of the stars - revolutions of the Sun God (Surya) = civil days per Mahayuga

1,582,237,828 - 4,320,000 = 1,577,917,828 civil days per Mahayuga

40. The intercalary months, the omitted lunar days, the risings of the stars (nakshatra), lunar, and civil days, these (of a Mahayuga), multiplied by 1000, are the number of revolutions, etc., in an Aeon (kalpa).

Comment:

1 kalpa = 1000 Mahayugas

41. The revolutions of the Sun's (Surya's) node of the apsis, moving eastward, in an Aeon (kalpa), are 387; of that of Mars (Mangal), 204; of that of Mercury (Budha), 368;

42. Of that of Jupiter (Guru), 900; of that of Venus (Sukra), 535; of the node of the apsis of Saturn (Shani), 39. Farther, the revolutions of the nodes of declination, retrograde, are:

43. Of that of Mars (Mangal), 214; of that of Mercury (Budha), 488; of that of Jupiter (Guru), 174; of that of Venus (Sukra), 903;

44. Of the node of declination of Saturn (Shani), the revolutions in an Aeon (kalpa) are 662; the revolutions of the moon's (Chandra's) node of the apsis and node of the declination have been given here already.

Comment: Revolutions of the node of the apsis, moving eastward, in an Aeon (kalpa); and revolutions of the node of the declination, retrograde, in an Aeon (kalpa):

apsis, declination

The Sun God (Surya) 387

Mars (Mangal) 204, 214

Mercury (Budha) 368, 448

Jupiter 900, 174

Venus 535, 903

Saturn (Shani) 39, 662

45. Now add together the time of the six Manu, with their respective twilights, and with the dawn at the start of the Aeon (kalpa); farther, of the Manu (Vaivasvata), son of Vivasvant,

Comment: Here is introduced a procedure referred to repeatedly, the sum of days (dinarasi). The sum of days can be made from the start of the creation, from the start of the finished creation, from the start of this manvantara, from the start of the last Silver Age (Treta yuga). Here the sum of days is to be from the start of the finished creation. Working from previous calculations:

Dawn of kalpa = 1,728,000

6 (manvantara + twilight) = 1,850,688,000

46. The 27 Ages (Mahayuga) that are past, and likewise the present Golden Age (Sathya Yuga); from their sum subtract the time of creation, already stated in terms of Divine years.

Comment: For the sum of days, working from previous calculations:

Dawn of kalpa = 1,728,000

6 (manvantara + twilight) = 1,850,688,000

27 Mahayugas of Vaivasvata = 116,640,000

Sathya Yuga = 1,728,000

Total = 1,970,784,000

Minus the time of creation = 17,064,000

Time from the start of the

finished creation to the

end of the Sathya Yuga = 1,953,720,000 solar years

47. In solar years: the result is the time elapsed at the end of the Golden Age (Sathya Yuga); namely, 1,953,720,000 solar years.

48. To this, add the number of years of the time since past. Reduce the sum to months, and add the months expired of the current year, beginning with the light half of Caithra.

Comment: Using the birth date of the avathara Sathya Sai Baba for the calculations of a sum of days: Sathya Sai Baba, November 23, 1926

Time from the start of the finished

creation till the end of the Sathya Yuga = 1,953,720,000

Treta Yuga = 1,296,000

Dwapara Yuga = 864,000

Sub total = 1,955,860,000

Kali Yuga,

from the spring equinox after

February 18, 3102 BC

till the spring equinox of 1926 AD = 5,027

Total = 1,955,865,027 solar years from the start of

the finished creation till the birth year of Sathya Sai Baba.

1,955,865,027 solar years x 12 = 23,470,380,324 solar months

Add the months expired of the current year. To do this we will need a calendar from India showing the intercalary month, if there is one, and the monthly progression for 1926. This would show the start of the month of Caithra. The month of Caithra is named after that month which has a full moon in the nakshatra Chithra, which is at Spica in Virgo (Kanya). Chithra (Spica) is nearly 180 degrees from the start of Aries (Mesha) at 10' east of Revati (Zeta Pisces). The lunar months are divided into light half (sukla paksha), which is new moon to full moon as the waxing half, and dark half (krishna paksha), which is full moon to new moon as the waning half. One would then count the months from the light half of Caithra, 1926, to November 26, 1926.

49. Set the result down in two places; multiply it by the number of intercalary months,

and divide by that of solar months, and add to the last result the number of intercalary months thus found; reduce the sum to days, and add the days expired of the current month;

Comment: Cross multiplying from the ratio:

Intercalary months in a Mahayuga / solar months in a Mahayuga =

intercalary months till November, 1926 / solar months till November, 1926

Solar months till November, 1926 x intercalary months /

solar months in a Mahayuga = intercalary months till November, 1926

Solar months till Novermber, 1926 + intercalary months till November, 1926

= lunar months till November, 1926

Lunar months till November, 1926 x 30 = lunar days till November, 1926

Lunar days till November, 1926 + lunar days elapsed of the last lunar month =

total lunar days (with omitted lunar days) from the start of the finished creation till November 23, 1926

50. Set the result down in two places; multiply it by the number of omitted lunar days, and divide by that of lunar days, subtract from the last result the number of omitted lunar days thus obtained: the remainder is, at midnight, on the meridian of Lanka (Ujjain, India),

Comment: For the sum of days till November, 1926. The math is from the ratio:

Lunar days in a Mahayuga / omitted days in a Mahayuga =

lunar days from the start of the finished creation till November 23, 1926 /

omitted lunar days from the start of the finished creation till November 23, 1926

lunar days from the start of the finished creation till November 23, 1926 -

omitted lunar days from the start of the finished creation till November 23, 1926 =

civil days from the start of the finished creation till November 23, 1926

This last result is called the dinarasi, the sum of days. In review, the calculations were: total solar years, minus the solar years for creating, solar months, plus intercalary months, lunar

months, lunar days, subtract the omitted lunar days, the result is the sum of civil days. This then is the dinarasi, the sum of days, in whole civil days and a remainder. It would take further math to see how the remainder is at midnight on the prime meridian of Ujjain, India. The prime meridian is presented in Sloka 62 of this chapter.

51. The sum of days, in civil reckoning, from this may be found the Lords of the day, the month, and the year, counting from the Sun (Surya). If the number be divided by 7, the remainder marks the Lord of the day, beginning with the Sun (Surya).

Comment: Given the sum of days, dividing it by 7 for whole cycles, and then looking at the remainder for how far the sum of days is into the last cycle. The list of the succession of the Lords of the day that the remainder marks is:

1. the Sun God (Surya)

2. the moon (Chandra)

3. Mars (Mangala)

4. Mercury (Budha)

5. Jupiter (Guru)

6. Venus (Sukra)

7. Saturn (Shani)

This would be the first 7 days from the start of the finished creation, the list repeats till the end of the sum of days.

The basic pattern for the Lords of the day is related to the Lords of the hour. Tradition in India orders the planets for the Lord of the hour as: Saturn (Shani), Jupiter (Guru), Mars (Mangal), the Sun God (Surya), Venus (Sukra), Mercury (Budha), the moon (Chandra). Using this order, and starting in the middle at the Sun God (Surya), and 24 hours, we have for the succession of the Lords of the hour:

The Lord of the first hour, of the first day, of the first year, of the finished creation is:

1. the Sun God (Surya) followed by

2. Venus (Sukra)

3. Mercury (Budha)

4. moon (Chandra).

5. Saturn (Shani)

6. Jupiter (Guru)

7. Mars (Mangal)

8. the Sun God (Surya)

9. Venus (Sukra)

10. Mercury (Budha)

11. moon (Chandra).

12. Saturn (Shani)

13. Jupiter (Guru)

14. Mars (Mangal)

15. the Sun God (Surya)

16. Venus (Sukra)

17. Mercury (Budha)

18. moon (Chandra).

19. Saturn (Shani)

20. Jupiter (Guru)

21. Mars (Mangal)

22. the Sun God (Surya)

23. Venus (Sukra)

24. Mercury (Budha)

The second day would start with 25. the moon (Chandra)

The third day would start with 49. Mars (Mangal)

The fourth day would start with 73. Mercury (Budha)

The fifth day would start with 97. Jupiter (Guru)

The sixth day would start with 121. Venus (Sukra)

The seventh day would start with 147. Saturn (Shani)

The Lord of the first hour of the day is the Lord of that day. This system is used also in the west to determine the same thing. From the Lord of the day we have the name of the day of the week. We then have the Lords of the 7 days, with their Greek names, their Indian names, the names of the days of the week in India, the Viking names of the planets, and the days of the week in the west:

1. The Sun God, Apollo Surya or Ravi Ravivara Sol Sunday

2. Moon goddess, Diana Chandra or Soma Somavara Mona Monday

3. Mars Mangal Mangalavara Thew Tuesday

4. Mercury Budha Budhavara Wodan Wednesday

5. Jupiter Guru Guruvara Thor Thursday

6. Venus Sukra Sukravara Frey Friday

7. Saturn Shani Shanivara Saturn Saturday

Diana is not so much the moon as the goddess of the moon.

One uses the above list for the Surya Siddhanta's statement: If the number be divided by 7, the remainder marks the Lord of the day, beginning with the Sun God (Surya). Example: Who is the Lord of the 12'th day? 12 / 7 = 1 remainder 5. The 5 spot on the above list is Jupiter (Guru) as Lord of the 12'th day.

The hourly progression continues from the first 7 days through the first 30 and the start of the 31'st, and so on. The Lord of the first hour of the first day of a month is the Lord of that hour and that day and that month.

The Lord of the first day and of the first month is the Sun God (Surya).

The Lord of the 2'nd month can be found by: 31 days / 7 = 4 remainder 3. The 3'rd Lord on the list is: Mars (Mangal) as Lord of the 2'nd month which would start on the 3'rd day of the week, Tuesday (Mangalavara).

The list for the progression of the Lords of the months is then:

1. the Sun God (Surya)

2. Mars (Mangal)

3. Jupiter (Guru)

4. Saturn (Shani)

5. the moon (Chandra)

6. Mercury (Budha)

7. Venus (Sukra)

The progression of the Lords of the day, with the Lords of the hours, and on out to the Lords of the month, continues to the Lords of the years. The years are 360 days each, one after another, so they coincide with the civil days only, not the solar years.

The Lord of the second years is then: 361 / 7 = 51 remainder 4. The 4'th Lord on the list for Lords of the day is: Mercury (Budha) as Lord of the second year which would start on the 4'th day of the week, Wednesday (Budhavara).

The list for the progression of the Lords of the years is then:

1. the Sun God (Surya)

2. Mercury (Budha)

3. Saturn (Shani)

4. Mars (Mangala)

5. Venus (Sukra)

6. the moon (Chandra)

7. Jupiter (Saturn)

52. Divide the same number by the number of days in a month and in a year, multiply the one quotient by 2 and the other by 3, add 1 to each product, and divide by 7; the remainders indicate the Lords of the month and of the year.

Comment: Comparing our lists of progression we find:

Lords of the day Lords of the month Lords of the year

1. the Sun God (Surya) 1. the Sun God (Surya) 1. the Sun God (Surya)

2. the moon (Chandra)

3. Mars (Mangala) 2. Mars (Mangala)

4. Mercury (Budha) 2. Mercury (Budha)

5. Jupiter (Guru) 3. Jupiter (Guru)

6. Venus (Sukra)

7. Saturn (Shani) 4. Saturn (Shani) 3. Saturn (Shani)

8. the Sun God (Surya)

9. the moon (Chandra) 5. the moon (Chandra)

10. Mars (Mangala) 4. Mars (Mangala)

11. Mercury (Budha) 6. Mercury (Budha)

12. Jupiter (Guru)

13. Venus (Sukra) 7. Venus (Sukra) 5. Venus (Sukra)

14. Saturn (Shani)

15. the Sun God (Surya) 8. the Sun God (Surya)

16. the moon (Chandra) 6. the moon (Chandra)

17. Mars (Mangala) 9. Mars (Mangala)

18. Mercury (Budha)

19. Jupiter (Guru) 10. Jupiter (Guru) 7. Jupiter (Guru)

20. Venus (Sukra)

21. Saturn (Shani) 11. Saturn (Shani)

The lists continue. It can be seen that the Lord of the day is each one in succession. The Lord of the month is every other one in succession. The Lord of the year is every third one in succession. From this it can be worked out:

Number of days / 7 = whole cycles of days + remainder

((number of days / 30 days per month) x 2) + 1 = whole cycles of months + remainder

((number of days / 360 days per year) x 3) + 1 = whole cycles of years + remainder

In each case the remainder indicates the placement on the list of succession.

53. Multiply the sum of days (dinarasi) by the number of revolutions of any planet, and divide by the number of civil days; the result is the position of that planet, in virtue of its mean motion, in revolutions and parts of a revolution.

Comment: The formula is from cross multiplying the ratio:

Number of civil days in a Mahayuga /

number of revolutions in a Mahayuga =

Sum of days from the start of the finished creation /

revolutions and parts of a revolution from the start of the finished creation

54. Thus also are ascertained the places of the node of the conjunction (sighra) and of the node of the apsis (mandocca) of each planet, which have been mentioned as moving eastward; and in like manner of the node of the declination, which have a retrograde motion, subtracting the result from a whole circle.

Comment: In a like manner one could cross multiply from the ratio:

Number of civil days in a Mahayuga /

number of revolutions of a node in a Mahayuga =

Sum of days from the start of the finished creation /

revolutions and parts of a revolution of a node from the start of the finished creation

The second part is from:

Western longitude + remainder of circle = 1 circle, where remainder of circle is the measure of eastern longitude.

55. Multiply by 12 the past revolutions of Jupiter (Guru), add the signs (rasi) of the current revolution, and divide by 60; the remainder marks the year of Jupiter's (Guru's) cycle, counting from Vijaya.

Comment:

((12 signs x the past revolutions of Jupiter) + signs of the current revolution) / 60 =

the whole number of Jupiter's (Guru's) cycles of 60 + remainder,

where the remainder indicates which year of the current cycle.

Jupiter as Guru is Jupiter as teacher. The students simply study the cycles of the teacher. It takes Jupiter (Guru) a little less than 1 solar year to travel through 1 sign of the zodiac, less than 12 years to travel through all 12 signs of the zodiac, and 5 full transits at less than 60 years to travel through 60 signs of the zodiac, which is called the cycle of Jupiter (Guru). Even though the cycle is counted from Vijaya, it is numbered the 27'th. The 60 years of the cycle of Jupiter are:

27. Vijaya	47. Pramadin	7. Shrimukha
28. Jaya	48. Ananda	8. Bhava
29. Manmatha	49. Rakshasa	9. Yuvan
30. Durmukha	50. Anala	10. Dhatar
31. Hemalamba	51. Pingala	11. Ishvara
32. Vilamba	52. Kalayukta	12. Bahudhanya
33. Vikarin	53. Siddharthin	13. Pramathin
34. Sharvari	54. Raudra	14. Vikrama
35. Plava	55. Durmati	15. Bhrishya
36. Shubhakrit	56. Dundubhi	16. Chithrabhanu
37. Shobhana	57. Rudhirodgarin	17. Subhanu
38. Krodhin	58. Raktaksha	18. Tarana
39. Vishvavasu	59. Krodhana	19. Parthiva
40. Parabhava	60. Kshaya	20. Vyaya
41. Plavanga	1. Prabhava	21. Sarvajit
42. Kilaka	2. Vibhava	22. Sarvadharin
43. Saumya	3. Sukla	23. Virodhin
44. Sadharana	4. Pramoda	24. Vikrita
45. Virodhakrit	5. Prajapati	25. Khara
46. Paridhavin	6. Angiras	26. Nandana

56. The processes which have thus been stated in full detail, are practically applied in an abridged form. The calculations of the mean place of the planets may be made from any epoch (yuga) that may be fixed upon.

57. Now, at the end of the Golden Age (Krita Yuga), all the planets, by their mean motion, excepting, however, their nodes of the apsis (mandocca) and their nodes of the declination, are in conjunction in the first of Aries (Mesha).

58. The moon's (Chandra's) node of the apsis (ucca) is in the first of Capricorn (Makara), and its node of the declination is in the first of Libra (Thula); and the rest have come to a position which is without a remainder of degrees; they being of slow motion are not stated here.

Comment: As seen from the earth, the moon and the planets would all align in front of and behind the sun in the first of Aries (Mesha). The end of the last Golden Age (Sathya Yuga) was 2,165,104 years ago. One can start the sum of days from then till the day in question. The end of the next Golden Age (Sathya Yuga) will be after the rest of this Iron Age (Kali Yuga) 426,896 more years, and all of the following Golden Age (Sathya Yuga) 1,728,000 years, for a total of 2,154,896 years from now.

The end of the last Golden Age (Sathya yuga):

Start of Aries (Mesha) the Sun God (Surya), the moon (Chandra), all the planets

Taurus (Vrishaba)

Gemini (Mithuna)

Cancer (Kakata)

Leo (Simha)

Virgo (Kanya)

start of Libra (Thula) the moon’s node of the declination

Scorpio (Vrischika)

Sagittarius (Dhanus)

start of Capricorn (Makara) the moon’s node of the apsis

Aquarius (Kumbha)

Pisces (Mina)

59. Twice 800 yojanas are the diameter of the earth (Bhumi): the square root of 10 times the square of that is the earth's (Bhumi's) circumference.

Comment: This brings to mind an equation that is different than 2 pi r = c, and a very different result:

((2 r) squared) x (square root 10) = circumference

((2 x 800) squared) x (square root 10) = 8,095,430.81... yojanas. A yojana is comparable to a league.

This would make the radius of the earth (Bhumi) about 1,280,000 yojanas, part way to Venus. Already my understanding is challenged.

We must remember that the Vedanta writes of the 7 earths. Theosophy and alchemy have also written and diagrammed the 7 earths. One could draw a vertical. This is the old symbols for 1 and for Divine origins. From 1, is 10, 100, 1000 and from 10 the square root of 10. One could then draw a circle around the vertical, put a horizontal diameter across the circle, for the cross of the earth (Bhumi), an old symbol for the four directions. One could label the horizontal diameter 2 x 800 yojanas, numbers proper to sacred geometry for the earth (Bhumi). These numbers also relate yojanas, which are like a league, or a country mile, to universal design.

Looking at the phrase "the square of the that" from the preceding sloka one would put a square circumscribing the circle such that the midpoints of the square are at the ends of the diameters. This is representative of Brahma Loka, that spiritual city of the creator characterized by a square. The statues of India depict the image of Brahma as being four headed. Out of the many arrangements of sacred geometry there is one that has been used in India and the surrounds for quite length of time. It is the one that has the two equal lateral triangles, one pointing down, one pointing up; surrounded by a circle and a second circle that is slightly larger, surrounded by an 8 petaled lotus, a 16 petaled lotus, a square, and geometry for the 2 and 3 parts of each of the four directions.

This geometry, the circle with diameters and circumscribing square, was used in the creating of the heavens and the earth and is used in the drawing of the gnomon circle, at Chapter III., sloka 5. A gnomon circle is an astronomical sundial.

The 7 earths (Bhumi) would then go out to mid heaven (Bhuva), part way to Venus (Sukra), and above the north pole to high heaven (Suvaha) where is Brahma Loka and the Gods. In this Siddhanta the earth's (Bhumi's) sphere is said to be in the middle of the golden egg of Brahma.

After looking at: ((2 r) squared) x (square root 10) = circumference, then looking at:

2 x (square root 10) x r = circumference

Square root 10 in an equation for circumference? 2 pi r = the perimeter of the area of a circle. A circumference is the distance around a circle. This suggests that the circumference must be greater than the perimeter. The square root of 10, about 3.162..., is greater than pi, about 3.14159... With 2 (square root 10) r = c, this yields a circumference that goes around the earth at a distance above its perimeter. Much like the length of a belt being longer than the perimeter of the waist.

2 x (square root 10) x 800 = 5059.644... yojanas in circumference

2 x pi x 800 = 5026.548... yojanas in perimeter

These with the previous calculation of 8,095,430.81... yojanas leave a person wondering how much they know about sacred geometry and universal design. From theosophy C. W. Leadbeater, and subsequently A. E. Powell write that the circumference of the astral earth (Bhumi) is between the perigee and apogee of the moon's orbit. Ultimately, for 7 earth's one would expect 7 circumferences.

60. This, multiplied by the lambajya (similar but different than a sine) of the latitude of any place, and divided by the radius, is the corrected circumference of the earth (Bhumi) at that place. Multiply the daily motion of a planet by the distance in longitude (decantara) of any place and divide by its corrected circumference;

Comment: Earth longitude is measured from one or both of two references. One is the prime meridian at Ujjain, India, presented in Sloka 62 of this chapter. The other is the due south meridian from the place of the observer. The astronomer uses a corrected circumference of the earth for the latitude of observation. Then the astronomer calculates the position of the planet relative to the prime meridian. Then the astronomer calculates the position of the planet relative to the place of observation.

The equation in terms of a lambajya? Lambajya is beyond my experience and education. However I think the basic ratios would be:

Radius of the earth / the lambajya of the latitude =

The circumference of the earth / corrected circumference of the circle of latitude

The equation, in terms of square root 10, implies that universal design is understood down to the earth where distance is measured in yojanas. Such are the Hindu astronomers.

The equation, in terms of cosine, can be done with trigonometry but it yields a different result..

Radius of the earth / the cosine of the latitude =

The circumference of the earth / circumference of the circle of latitude

The astronomer calculates the place of the planet, this gives the position of the planet relative to the prime meridian. Then the astronomer calculates the position of the planet relative to the place of observation.

Longitude from the prime meridian / circumference of the circle of latitude =

Motion of planet from prime meridian to over the observer / daily motion of planet

61. The quotient, in minutes, subtract from the mean position of the planet as found, if the place be east of the prime meridian (rekha); add if it be west; the result is the planet's mean position at the given place.

Comment: For an observer east of the prime meridian of Ujjain::

Longitude of the position of the planet at Ujjain - longitude of the observer =

Longitude of the position of the planet at the place of observation.

For an observer west of Ujjain:

Longitude of the position of the planet at Ujjain + longitude of the observer =

Longitude of the position of the planet at the place of observation.

62. Situated upon the line which passes through the haunt of the rakshasas and the mountain which is the seat of the Gods (suras), are Rohitaka and Avanti, as also the adjacent lake.

Comment: Rohitaka is said to be a city in India during the times of Atlantis. Avanti is said to be an old name for Ujjain, India and is regarded as the city of the prime meridian, both during Atlantis and now. This would place Greenwich at 75 degrees, 47' west longitude from the prime meridian of Ujjain. Rakshasas are one of the lesser type of people. The mountain which is the seat of the Gods is Mount Meru. It rises above the north pole from the earth (Bhumi), through mid heaven (Bhuva) where is the pole star, to high heaven (Suvaha) where is Brahmaloka (world of Brahma). Mt. Meru also extends below the south pole, through mid heaven, to high heaven. This constitutes the central axis of a Divinely illumined geocentric creation (Brahma). Around this axis is the revolution of the ether carrying the stars as the first of three hoops, the hoop of the lunar moon mansions of Chandra (nakshatra), and the hoop of the ecliptic where is the Sun God (Surya) and the path of the planets. A great circle above and below the north and south pole, Mount Meru, would have a corresponding meridian on the earth. This meridian passes through Ujjain, India, and has been considered the prime

meridian since the time of this Siddhanta, or indeed, even before, as it is part of heaven earth design. From Brahmaloka to Siddha, the city of the north; to Rohitaka on the prime meridian in the south; to Yamakoti, 90 degrees east; to Romaka, 90 degrees west; to Patala, 'the other side,' 180 degrees east. One notes that India is, therefore, south and it is China that is east, West Africa as west, and of course Mexico and Peru are Patala.

63. When, in a total eclipse of the moon, the emergence (unmilana) takes place after the calculated time for its occurrence, then the place of the observer is to the east of the central meridian;

Comment: Unmilana refers to the moon's leading edge just entering the earth's shadow. The calculated time for its occurrence is in reference to the prime meridian. Time is later to the east and earlier to the west. An eclipse that occurs at midnight in Ujjain would be seen as occurring near dawn at 90 degrees east and near eve at 90 degrees west.

64. When it takes place before the calculated time, his place is to the west: the same thing may be ascertained likewise from the immersion (nimilana). Multiply by the difference of the two times in nadis the corrected circumference of the earth (Bhumi) at the place of observation,

Comment: Nimilana refers to the moon's leading edge leaving the earth's shadow..

Time in nadis / 60 nadis in 1 day of respirations =

distance from Ujjain / corrected circumference of the circle of latitude

65. And divide by sixty: the result, in yojanas, indicates the distance of the observer from the meridian, to the east or to the west, upon his own parallel; and by means of that is made the correction for difference of longitude.

Comment: Cross multiplying the previous ratio::

(Time in nadis x corrected circumference) / 60 nadis = distance from Ujjain

66. The succession of the week day (vara) takes place, to the east of the meridian, at a time after midnight equal to the difference of longitude in nadis; to the west of the meridian, at a corresponding time before midnight.

Comment::

Time at Ujjain + longitude in nadis = time east of Ujjain

Time at Ujjain - longitude in nadis = time west of Ujjain

67. Multiply the mean daily motion of a planet by the number of nadis of the time fixed

upon, and divide by sixty: subtract the quotient from the place of the planet, if the time be before midnight; add, if it be after: the result is its place at the given time.

Comment: The formula is from cross multiplying the ratio::

Nadis between midnight and the given time / 60 nadis in 1 day of respirations =

motion between midnight and the given time / motion in 1 day

The place of the planet in revolutions at midnight -

the place of the planet between midnight and the given time =

the place of the planet at the given time before midnight

The place of the planet in revolutions at midnight +

the place of the planet between midnight and the given time =

the place of the planet at the given time after midnight

68. The moon (Chandra) is, by its node, caused to deviate from the limit of its declination (kranti), northward and southward, to a distance, when greatest, of an eightieth part of the minutes of a circle;

Comment: Declination is a planet's movement above, north of; or below, south of; the ecliptic. Longitude from either the start of Aries (Mesha) or from the prime meridian of Ujjain or from the southern meridian of the place of observation, with declination as north or south of the ecliptic, determine a planet's location. The moon's declination, when greatest, is the eightieth part of the minutes of a circle:

21,600 minutes in a circle / 80 = 270' = 4 degrees 30', moon's greatest declination

1/80 + 79/80 = 1 circle

270 + 21,330 = 21,600 minutes in a circle 21,330 / 21,600 = 79 / 80

600 + 47,400 = 48,000 Divine years 47,400 / 48,000 = 79 / 80

69. Jupiter (Guru), to the 9'th part of that multiplied by 2; Mars (Mangal), to the same amount multiplied by 3; Mercury (Budha), Venus (Sukra), and Saturn (Shani) are by their nodes caused to deviate to the same amount multiplied by 4.

Comment: Well, I'm inclined to say this is the declinations of the planets::

Jupiter (Guru) (270' / 9) x 2 = 60' = 1 degree

Mars (Mangal) (270’ / 9) x 3 = 90’ = 1 degree 30’

Mercury (Budha) (270’ / 9) x 4 = 120’ = 2 degrees

Venus (Sukra) (270’ / 9) x 4 = 120’ = 2 degrees

Saturn (Shani) (270’ / 9) x 4 = 120’ = 2 degrees

70. So also, 27, 9, 12, 6, 12 and 12 multiplied respectively by 10, give the number of minutes of mean latitude (vikshepa) of the moon (Chandra) and the rest, in their order.

Comment:

10 from the 1 increasing:

3 cubed 27

3 squared 9

2 squared 3 12

2 times 3 6

2 squared 3 12

2 squared 3 12

Declinations of the planets:

the moon (Chandra) 27 x 10 = 270’ = 4 degrees 30’

Mars (Mangal) 9 x 10 = 90’ = 1 degree 30’

Mercury (Budha) 12 x 10 = 120’ = 2 degrees

Jupiter (Guru) 6 x 10 = 60’ = 1 degree

Venus (Sukra) 12 x 10 = 120’ = 2 degrees

Saturn (Shani) 12 x 10 = 120’ = 2 degrees

Comment: If one has a feeling for Pythagorean number and universal design this last verse of Chapter I is profound although hard to translate from numbers to words.

I need to read it again.

Alaska Mark at www .theplatonicsolid. com

- Complete Table of Contents, long form, contains a description of each page
- Partial Table of Contents, short form, at the bottom of this page

- Madam Blavatsky on the Surya Siddhanta dry and serious

- Surya Siddhanta, Chapter I, the Motions of the Planets, just the text, no calculations difficult

- This page, Surya Siddhanta, Chapter I, the Motions of the Planets, with Comments and Calculations, important

- Next page, Shirdi Sai, the Peace Maker, stories from his life and times

- Top of Page

(Page report. I seem to not only have diagrammed the 'coils of Maya' but I seem to be entangled therein. Unfinished, the diagramming and the disentangling continues.)

(Development of theme within site mission. This page establishes **the Surya Siddhanta as the central tradition in chronology.** The theme progresses from Pythagorean number, to Pythagoras, Greek use of Pythagorean number, history of numerical tradition in mystery schools, the works of Plato in the Academy, the eternal soul as remembering what it already knows, the role of pattern recognition in sacred geometry as an aid in the memory of the eternal soul, the study of Platonic solids in the development of the higher mental, **the study of Eastern wisdom in the final development of spiritual ability,** the progression from number to geometry to solid geometry to **astronomy** to music to self realization, an examination of the aura as a visual indication of the progression of the individual, the appearance of the Platonic solids in the human aura, observation of atoms with meditative procedure, the appearance of the Platonic solids in the atom, Pythagorean number and Platonic solids as evident in Genesis, the Jesus went to India story to establish the connection between the Hebrew and Hindu traditions, the emergence of abilities from the development of the higher mental, observation of the septenary Earth, the Platonic solids in the Earth grid, the proper use of the higher abilities, the illusory nature of the higher abilities, a comparison of views of future events, the illusory appearances in time, the focusing of the attention on Aum and eternity.)

The 7 stars of the Big Dipper are the 7 stars of the 7 Rishis.

Mizar and Alcor are two stars along the same line of sight. Alcor is used as a test for eyesight.

The 'toes' of the Bear are called 'the three leaps of the gazelle.'

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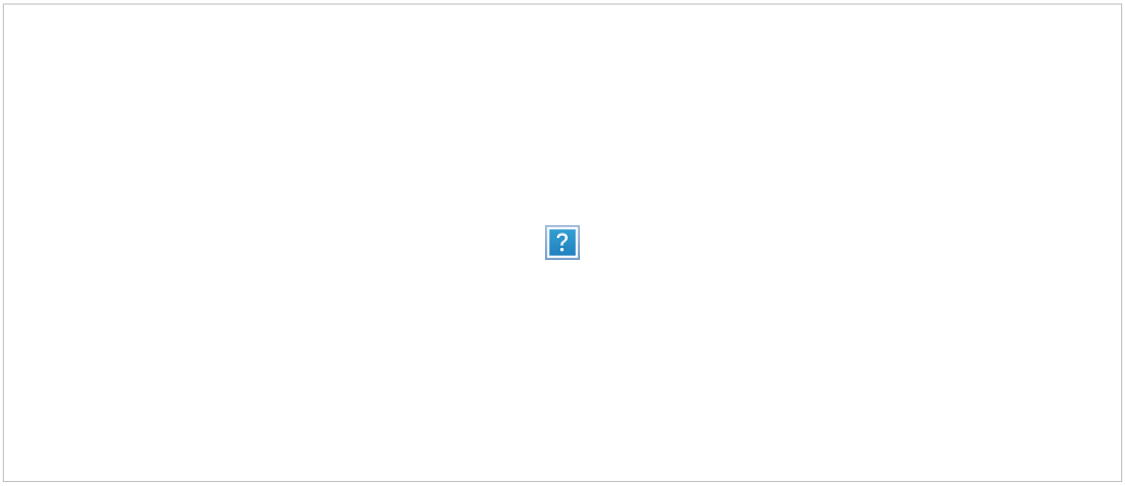

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